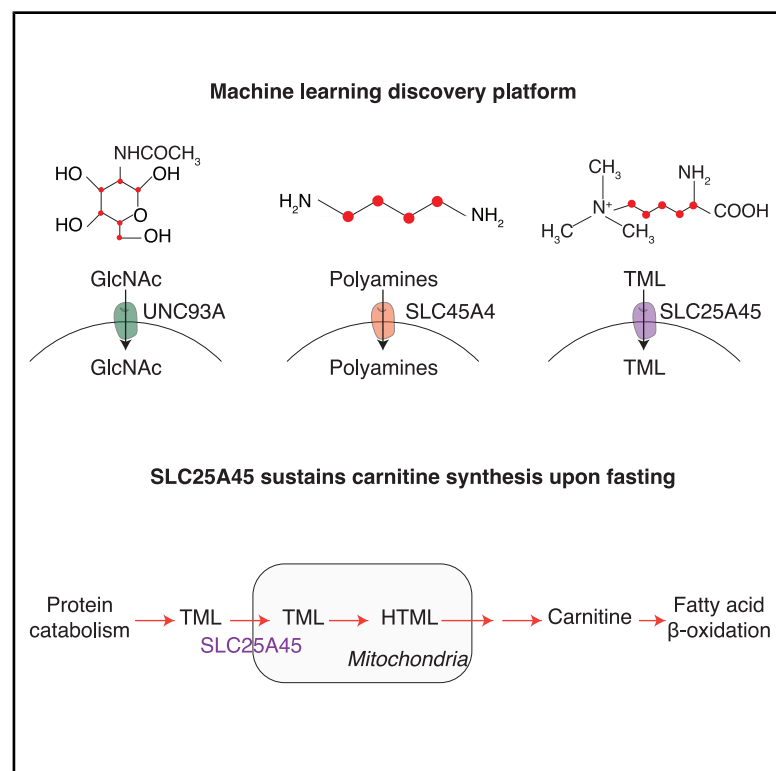


Cell Metabolism

Machine-learning-guided discovery of SLC25A45 as a mediator of mitochondrial methylated amino acid import and carnitine synthesis

Graphical abstract



Authors

Artem Khan, Frederick S. Yen, Gokhan Unlu, ..., Eric R. Gamazon, Gary J. Patti, Kivanç Birsoy

Correspondence

kbirsoy@rockefeller.edu

In brief

Khan et al. apply machine learning to identify substrates of UNC93A, SLC45A4, and SLC25A45. Biochemical assays and *in vivo* metabolite tracing establish SLC25A45 as a key mediator of mitochondrial import of methylated amino acids and carnitine synthesis. Loss of *Slc25a45* impairs fasting response by limiting carnitine availability for β -oxidation.

Highlights

- Machine-learning platform predicts and prioritizes physiological substrates for SLCs
- UNC93A and SLC45A4 mediate transport of acetylglucosamine and polyamines, respectively
- Loss of *Slc25a45* blunts fasting response by impairing carnitine synthesis
- SLC25A45 mediates mitochondrial import of methylated amino acids



Article

Machine-learning-guided discovery of SLC25A45 as a mediator of mitochondrial methylated amino acid import and carnitine synthesis

Artem Khan,¹ Frederick S. Yen,^{1,5} Gokhan Unlu,^{1,5} Nicole L. DelGaudio,^{1,5} Ranya Erdal,¹ Michael Xiao,¹ Khando Wangdu,¹ Kevin Cho,² Eric R. Gamazon,^{3,4} Gary J. Patti,² and Kivanç Birsoy^{1,6,*}

¹Laboratory of Metabolic Regulation and Genetics, The Rockefeller University, New York, NY, USA

²Department of Chemistry, Genetics, Medicine, and Center for Mass Spectrometry and Metabolic Tracing, Washington University, St. Louis, MO, USA

³Vanderbilt Diabetes Center, Vanderbilt University Medical Center, Nashville, TN, USA

⁴Vanderbilt Diabetes Research and Training Center, Vanderbilt University, Nashville, TN, USA

⁵These authors contributed equally

⁶Lead contact

*Correspondence: kbirsoy@rockefeller.edu

<https://doi.org/10.1016/j.cmet.2025.09.015>

SUMMARY

Solute carriers (SLCs) regulate cellular and organismal metabolism by transporting small molecules and ions across membranes, yet the physiological substrates of ~20% remain elusive. To address this, we developed a machine-learning platform to predict gene-metabolite associations. This approach identifies UNC93A and SLC45A4 as candidate plasma membrane transporters for acetylglucosamine and polyamines, respectively. Additionally, we uncover SLC25A45 as a mitochondrial transporter linked to serum levels of methylated basic amino acids, products of protein catabolism. Mechanistically, SLC25A45 is necessary for the mitochondrial import of methylated basic amino acids, including ADMA and TML, the latter serving as a precursor for carnitine synthesis. In line with this observation, SLC25A45 loss impairs carnitine synthesis and blunts upregulation of carnitine-containing metabolites under fasted conditions. By facilitating mitochondrial TML import, SLC25A45 connects protein catabolism to carnitine production, sustaining β -oxidation during fasting. Altogether, our study identifies putative substrates for three SLCs and provides a resource for transporter deorphanization.

INTRODUCTION

Solute carrier (SLC) transporters are central to metabolic processes, facilitating the translocation of small molecules and ions across membranes to shape the biochemical landscape. Dysfunction in metabolite transport is frequently implicated in disorders ranging from congenital metabolic diseases to cancer.¹ Despite the substantial progress in characterizing hundreds of SLCs, the molecular substrates of many remain unknown.^{1,2} This is largely due to their diverse tissue-expression patterns, functional redundancy, and substrate promiscuity.¹ Recent genome-wide association studies (GWASs) of the human serum metabolome address these challenges by leveraging circulating metabolites as a readout of cumulative metabolic activity across major organs.^{3,4} With thousands of samples and broad metabolite coverage (>1,000 metabolites), these studies offer high statistical power to detect even subtle effects in an unbiased manner.

GWAS-driven approaches have uncovered substrates of not only plasma membrane but also mitochondrial transporters,

emphasizing the influence of subcellular metabolism on the serum metabolome. Indeed, mitochondrial transport is essential for the synthesis of several metabolites by enabling the exchange of intermediates between mitochondria and the cytosol. A recent example of such discoveries is SLC25A48, whose association with serum choline levels led to its identification as a mitochondrial choline importer, filling a critical knowledge gap in the synthesis of betaine, a key metabolite in one-carbon metabolism.⁴ Similar metabolic coordination exists in carnitine biosynthesis, where precursor metabolites must enter mitochondria and undergo transformation for their production in the cytosol. Notably, these pathways are often compartmentalized across not only organelles but also organs, with some reactions confined to distinct tissues. This organismal and subcellular organization further complicates studying the molecular players involved in metabolic processes. For instance, carnitine synthesis occurs in kidney, liver, and brain,⁵ but mitochondrial transporters mediating key carnitine precursors such as trimethyllysine (TML) and hydroxytrimethyllysine (HTML) remain unknown.⁶ These knowledge gaps offer an opportunity for



GWAS-based approaches to resolve the spatial compartmentalization of these pathways and identify their associated transporters.

Despite the utility of GWASs, existing prioritization methods rely heavily on statistical filtering^{3,4} and do not fully leverage the extensive experimental metabolic pathway knowledge, underscoring the need for improved discovery strategies. To address this, we developed a supervised machine-learning framework trained on known gene-metabolite associations to predict and prioritize novel transport functions. Using this approach, we propose polyamines and acetylglucosamine as candidate substrates of SLC45A4 and UNC93A, respectively. Isotope-labeled tracing confirmed these predictions, demonstrating that SLC45A4 and UNC93A mediate the transport of polyamine and acetylglucosamine species across the plasma membrane. Beyond these plasma membrane transporters, we identified a genetic association between the orphan mitochondrial transporter SLC25A45 and serum methylated basic amino acids, which originate from protein catabolism. Among these amino acids, TML depends on SLC25A45-mediated transport for its metabolic fate in generating carnitine, a key metabolite required for β -oxidation and energy metabolism. SLC25A45 loss in mice impairs carnitine production and fatty acid oxidation, particularly under fasting conditions when these processes are most critical for energy homeostasis. Biochemical experiments revealed that SLC25A45 facilitates mitochondrial TML import in mammalian cells. Altogether, our work establishes SLC25A45-mediated mitochondrial import of methylated amino acids as a key link between protein catabolism and β -oxidation. Additionally, our study highlights the utility of computational approaches in advancing metabolic transporter function discovery and its implications for human health and disease.

RESULTS AND DISCUSSION

Machine learning predicts the function of metabolic genes

To fully leverage our understanding of empirical biochemical knowledge, we developed a supervised machine-learning approach trained on experimentally validated gene-metabolite pairs to predict novel associations. This model assigns a probability score to each gene-metabolite interaction, providing a systematic framework for prioritizing candidate transporters and enzymes. A key advantage of this approach is its automated, unbiased feature weighing, followed by ranking associations based on probabilistic scoring as opposed to conventional sequential, feature-based statistical cutoffs. Building on our previous success with transcriptome-wide association studies (TWASs)^{4,7} and GWASs in predicting physiological substrates of metabolic enzymes and transporters, we engineered features based on the METSIM⁸ and CLSA⁹ plasma metabolome GWAS datasets. Additionally, we incorporated gene-related properties, such as CADD pathogenicity scores,¹⁰ as well as biochemical and functional similarity of metabolites, to further enhance prediction accuracy and biological relevance. Following feature engineering, we trained and fine-tuned machine-learning models on a balanced dataset of 1,530 pairs that include known and randomly sampled gene-metabolite associations. The performance was assessed by 5-fold cross-validation and area under

the curve (AUC) as a metric for precision-recall balance (Figure 1A). Among the models tested, random forest (RF) demonstrated a marginally superior performance compared to gradient boosting (GB) and substantially outperformed neural networks and support vector machine (Figures 1B, 1C, S1A, and S1B). Notably, RF utilized all 17 features with equal weights (Figure S1C).

We therefore employed RF to compute probabilistic associations across 2,426,064 gene-metabolite pairs. Consistent with the expectation that most associations are not biologically relevant, the distribution of probabilistic scores was skewed toward lower values (Figures 1D and 1E). Remarkably, given that we did not explicitly incorporate gene function into the model, the predicted probability score distribution strongly correlated with known biological functions, particularly for SLC transporters (Figure 1F). Accordingly, gene-metabolite pairs involving metabolic genes were specifically enriched among associations with higher probability scores (Figure 1G). The entries with SLCs followed the same trend with respect to both all associations (Figure 1H) and associations with metabolic genes (Figure 1I). Building upon the robustness and biological relevance of machine-learning predictions, we prioritized 430 top-scoring associations involving 160 SLCs (Figures 1J and S1D). These associations form a network with well-defined and biochemically meaningful Louvain communities (Figure 1K). Among the prioritized associations, we found supporting evidence in the literature for 36 out of 160 SLCs with established functions (Figure 2A). For instance, we identify SLC7A6, a cationic amino acid transporter,^{11–13} associated with lysine metabolites. Additionally, our analysis links a lysophospholipid transporter, MFSD2A,¹⁴ to choline, phosphoethanolamine, oleate, and stearate. Altogether, these results indicate that our model captures bona fide biological relationships between genes and metabolites.

Substrate assignment for orphan solute carriers

We next sought to utilize our approach for uncovering novel substrates for transporters. For example, our analysis links the sugar transporter SLC45A4 to polyamine species that have not been recognized as substrates and cannot be explained by the known sugar transport (Figure 2A). To experimentally validate this association, we generated HEK293T SLC45A4-knockout cells expressing SLC45A4 complementary DNA (cDNA) or a vector control (Figures S2A–S2C) and conducted polar metabolite profiling using liquid chromatography-mass spectrometry (LC-MS). Consistent with the machine-learning predictions, acetylspermidine levels were depleted by nearly 4-fold, while gamma-aminobutyric acid (GABA) accumulated by $\sim$ 5.5-fold upon SLC45A4 expression (Figure 2B), implicating SLC45A4 in controlling the intracellular levels of polyamine species. Notably, a possible explanation for GABA accumulation is an extracellular conversion of putrescine to 4-aminobutanal (ABAL), which is imported back into the cells for GABA synthesis (Figure S2D).¹⁵ Indeed, isotope-labeled tracing with d_6 -spermidine and [$^{13}C_4$] putrescine showed a depletion of downstream metabolites, d_6 -acetylspermidine and [$^{13}C_4$]acetylputrescine, upon SLC45A4 expression, respectively (Figures 2C, 2D, S2E, and S2F). Furthermore, SLC45A4 expression resulted in elevated levels of [$^{13}C_4$]acetylputrescine in the media (Figure S2G). In line with these findings, recent work that conducted a series of structural

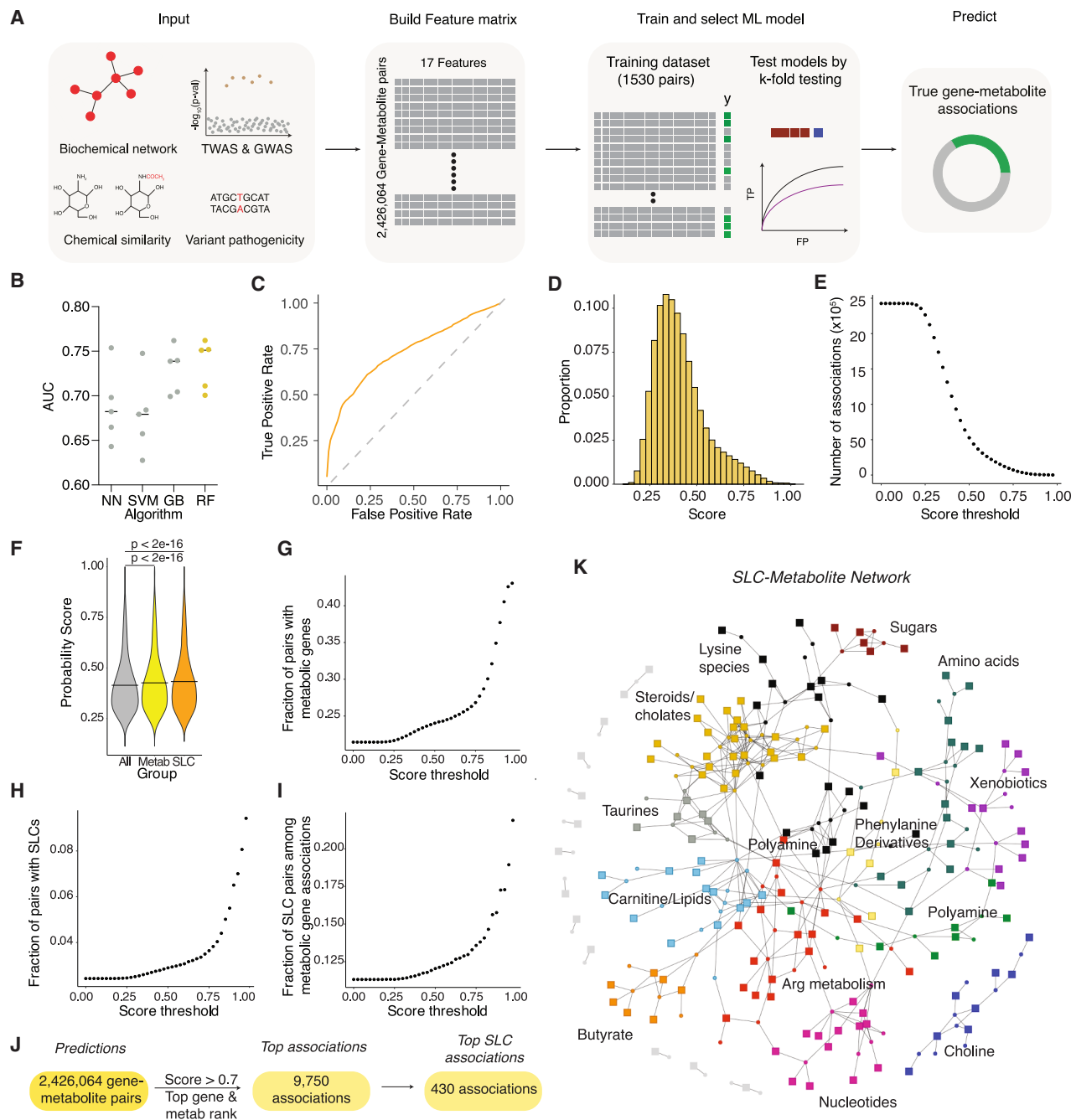


Figure 1. Machine-learning-driven prediction of the gene-metabolite associations

(A) Development of the supervised machine-learning pipeline for the prediction of gene-metabolite associations.

(B) Area under the curve (AUC) values from the 5-fold testing for the tested machine-learning algorithms. NN, neural network; SVM, support vector machine; GB, gradient boosting; RF, random forest.

(C) Average precision-recall curve for the random forest (orange). Dotted gray line displays $y = x$ line.

(D) Histogram displaying the distribution of probability scores generated from the machine-learning prediction pipeline.

(E) Number of the gene-metabolite pairs passing the probability score threshold.

(F) Violin plot showing the distribution of the probability scores for the pairs with all (gray), metabolic (yellow), and SLC-encoding genes (orange). The line represents the mean. Statistical significance was determined by Welch two-sample t test.

(G) Fraction of pairs with metabolic genes among all the associations passing the indicated threshold.

(H) Fraction of pairs with SLCs among all the associations passing the indicated threshold.

(I) Fraction of pairs with SLCs among the associations with metabolic genes passing the indicated threshold.

(J) Schematic showing filtering and prioritization of SLC-metabolite associations.

(K) SLC-metabolite network with the annotated Louvain communities.

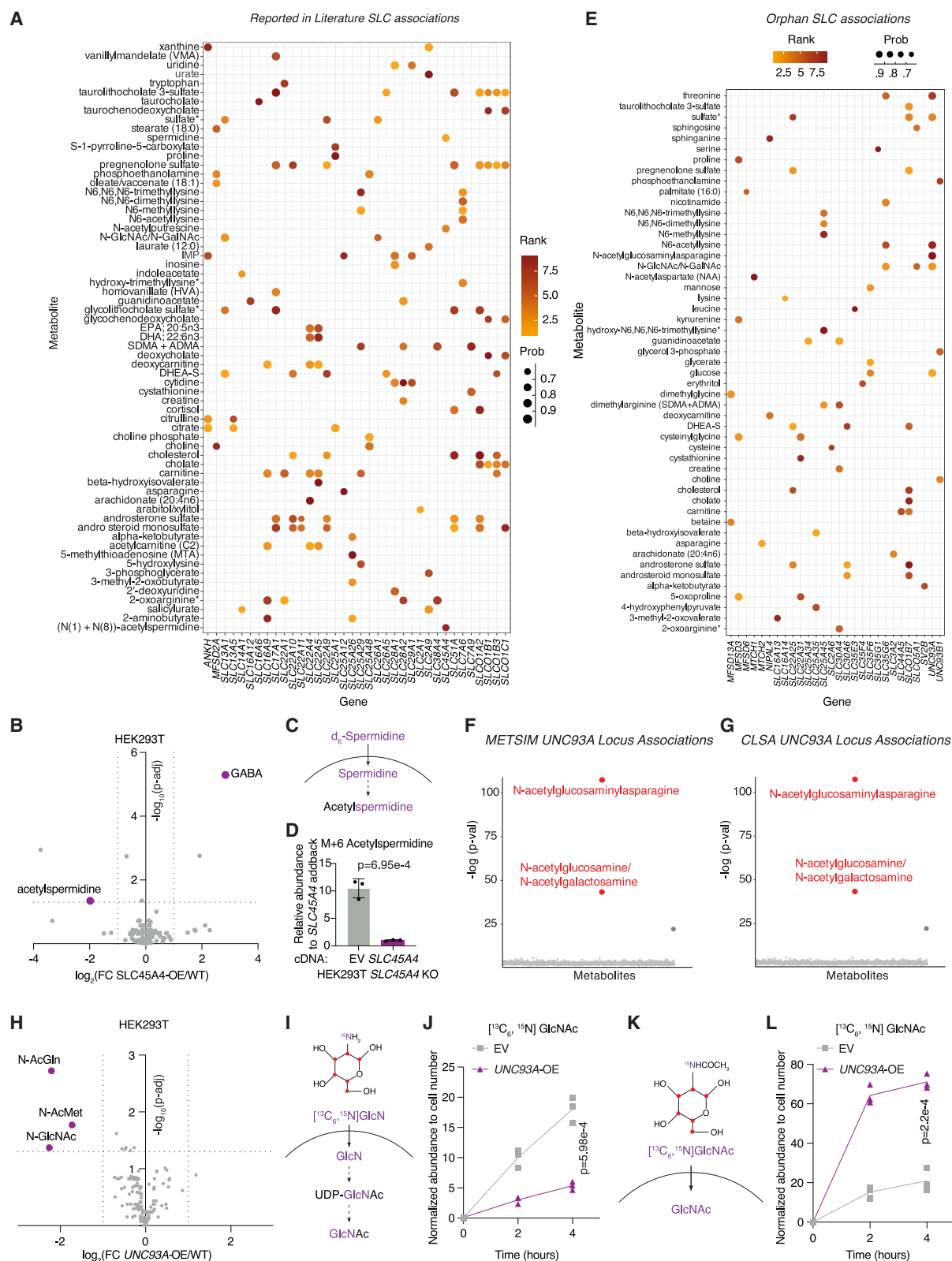


Figure 2. Machine learning identifies physiological substrates of solute carrier transporters

(A) Bubble plot of the SLC-metabolite associations with supporting literature evidence. Bubble color corresponds to the metabolic rank for the indicated gene. Bubble size represents the probability score of the association between the indicated gene and metabolite.

(legend continued on next page)

and biochemical assays identified SLC45A4 as a plasma membrane polyamine transporter.^{16,17}

We next rendered the list of associations with 28 orphan SLCs (Figure 2E) and focused on the pleiotropic genes (Figure S2H). Among those, we identified two genes outside of the paralogous loci: *SLC25A45* and *UNC93A*, linked to methylated basic amino acids and acetylglucosamine species, respectively (Figure S2H). In serum data, *UNC93A* locus was significantly associated with two acetylglucosamine species: acetylglucosamine (GlcNAc)/acetylgalactosamine (GalNAc) and GlcNAc-Asn (acetylglucosamine-asparagine) (Figures 2F and 2G).^{8,9} GlcNAc is involved in the hexosamine biosynthetic pathway, serving as a key nutrient sensor that regulates signaling and protein function.¹⁸ GlcNAc-Asn is a byproduct of glycoprotein degradation, released during lysosomal breakdown of glycoproteins.¹⁹ It is subsequently processed by specialized enzymes to recycle GlcNAc for cellular use.¹⁹ Furthermore, GlcNAc is a building block of hyaluronic acid, a critical glycosaminoglycan for skin health.^{20,21} Thus, in line with these known functions of GlcNAc, *UNC93A* has a restricted expression to the skin and liver²² with little to no expression in other tissues (Figure S2I). Therefore, to study its function, we overexpressed *UNC93A* cDNA in HEK293T cells (Figures S2J and S2K) and performed metabolite profiling. Consistent with its predicted localization, immunofluorescence analysis confirmed *UNC93A* localized at the plasma membrane (Figure S2L). Whereas most metabolites were not substantially altered by *UNC93A* overexpression, in accordance with the GWAS data, GlcNAc was depleted by over 8-fold (Figure 2H), raising the possibility that *UNC93A* can mediate cellular export of acetylglucosamine or relevant metabolites. To test this possibility, we conducted isotope-labeled [¹³C₆, ¹⁵N]glucosamine tracing (Figure 2I). We observed that the level of [¹³C₆, ¹⁵N]GlcNAc was indeed substantially decreased in *UNC93A*-overexpressing cells (Figure 2J), without impacting those of [¹³C₆, ¹⁵N]UDP-GlcNAc and [¹³C₆, ¹⁵N]glucosamine (Figures S2M and S2N). To further confirm this, we leveraged the characteristic property of SLCs as bidirectional transporters facilitating concentration gradient-dependent movement across

membranes and measured uptake of [¹³C₆, ¹⁵N]GlcNAc and its conversion into UDP-GlcNAc, a downstream product (Figure 2K). Strikingly, cells overexpressing *UNC93A* displayed a 3.5- and 10,000-fold increase in [¹³C₆, ¹⁵N]GlcNAc and [¹³C₆, ¹⁵N]UDP-GlcNAc, respectively (Figures 2L and S2O). We also tested the uptake of GlcNAc-Asn, another metabolite associated with *UNC93A* variants, and found that *UNC93A* overexpression similarly resulted in a 10-fold accumulation (Figures S2P and S2Q). Altogether, our data suggest that *UNC93A* mediates the transport of acetylated glucosamine species across the plasma membrane.

SLC25A45 is required for carnitine biosynthesis *in vivo*

Building on the success of our machine-learning approach for SLC function discovery, we focused on another association predicted by our platform: *SLC25A45* and methylated basic amino acids including TML, ADMA, and dimethyllysine (DML). Methylated amino acids are degradation products of proteins with posttranslational modifications and play important roles in metabolism and physiology. For example, ADMA acts as an endogenous inhibitor of nitric oxide (NO) synthesis, influencing cardiovascular health by modulating vascular tone.²³ TML is a precursor in the biosynthesis of carnitine, a critical metabolite in mitochondrial long-chain fatty acid β -oxidation. Although carnitine is predominantly acquired through diet, approximately 25% is synthesized endogenously,²⁴ highlighting the physiological importance of understanding this metabolic route. Indeed, in the METSIM⁸ study, one of the most significant associations at the *SLC25A45* locus was TML and deoxycarnitine, a downstream metabolic product of TML and precursor in carnitine synthesis (Figure 3A).

To investigate these predicted associations *in vivo*, we generated a whole-body *Slc25a45*-knockout mouse model using CRISPR (Figures S3A and S3B) and performed serum metabolite profiling relative to littermate wild-type controls (Figure 3B). Notably, to minimize exogenous carnitine intake prior to the experiment, mice were fed a vegetarian chow for 2 weeks prior to the experiment, which did not lead to any differences in

(B) Volcano plot with $-\log_{10}(p \text{ adj})$ versus $\log_2(\text{fold change})$ in metabolite abundance normalized to tryptophan from HEK293T *SLC45A4* overexpression and corresponding control cells. The horizontal dotted line is the significance threshold of $p \text{ adj} < 0.05$. The vertical dotted line corresponds to $\text{abs}(\log_2\text{FC}) = 1$. Two-tailed unpaired t tests followed by Benjamini-Hochberg multiple-test correction were performed; $n = 3$ biological replicates.

(C) Schematic for the cellular d₆-spermidine isotope tracing.

(D) d₆-acetylspermidine abundance in HEK293T *SLC45A4*-knockout cells carrying an empty vector control (gray) or *3xFlag-SLC45A4* cDNA (purple) after incubation with 10 μM d₆-spermidine for 1 h. Data are individual points; $n = 3$ biological replicates. Lines correspond to the mean for each time point. Statistical significance was determined by two-tailed unpaired t test.

(E) Bubble plot of the orphan SLC-metabolite associations. Bubble color corresponds to the metabolic rank for the indicated gene. Bubble size represents the probability score of the association between the indicated gene and metabolite.

(F and G) METSIM (F) and CLSA (G) *UNC93A*-metabolite associations displayed as $-\log(\min p \text{ value})$ for the given metabolite within the defined *UNC93A* locus vs. metabolites in alphabetical order.

(H) Volcano plot with $-\log_{10}(p \text{ adj})$ versus $\log_2(\text{fold change})$ in metabolite abundance normalized to tryptophan from HEK293T *UNC93A* overexpression and corresponding control cells. The horizontal dotted line is the significance threshold of $p \text{ adj} < 0.05$. The vertical dotted line corresponds to $\text{abs}(\log_2\text{FC}) = 1$. Two-tailed unpaired t tests followed by Benjamini-Hochberg multiple-test correction were performed; $n = 3$ biological replicates.

(I) Schematic for the cellular [¹³C₆, ¹⁵N]glucosamine (GlcN) isotope tracing.

(J) [¹³C₆, ¹⁵N]GlcNAc abundance in HEK293T cells carrying an empty vector control (gray) or *UNC93A-3xFlag* cDNA (purple) after incubation with 100 μM [¹³C₆, ¹⁵N]GlcN for the indicated time points. Data are individual points and were normalized to the number of cells seeded; $n = 3$ biological replicates. Lines correspond to the mean for each time point. Statistical significance was determined by two-tailed unpaired t test for the final time point.

(K) Schematic for the cellular [¹³C₆, ¹⁵N]GlcNAc isotope tracing.

(L) [¹³C₆, ¹⁵N]GlcNAc abundance in HEK293T cells carrying an empty vector control (gray) or *UNC93A-3xFlag* cDNA (purple) after incubation with 100 μM [¹³C₆, ¹⁵N]GlcNAc for the indicated time points. Data are individual points and were normalized to the number of cells seeded; $n = 3$ biological replicates. Lines correspond to the mean for each time point. Statistical significance was determined by two-tailed unpaired t test for the final time point.

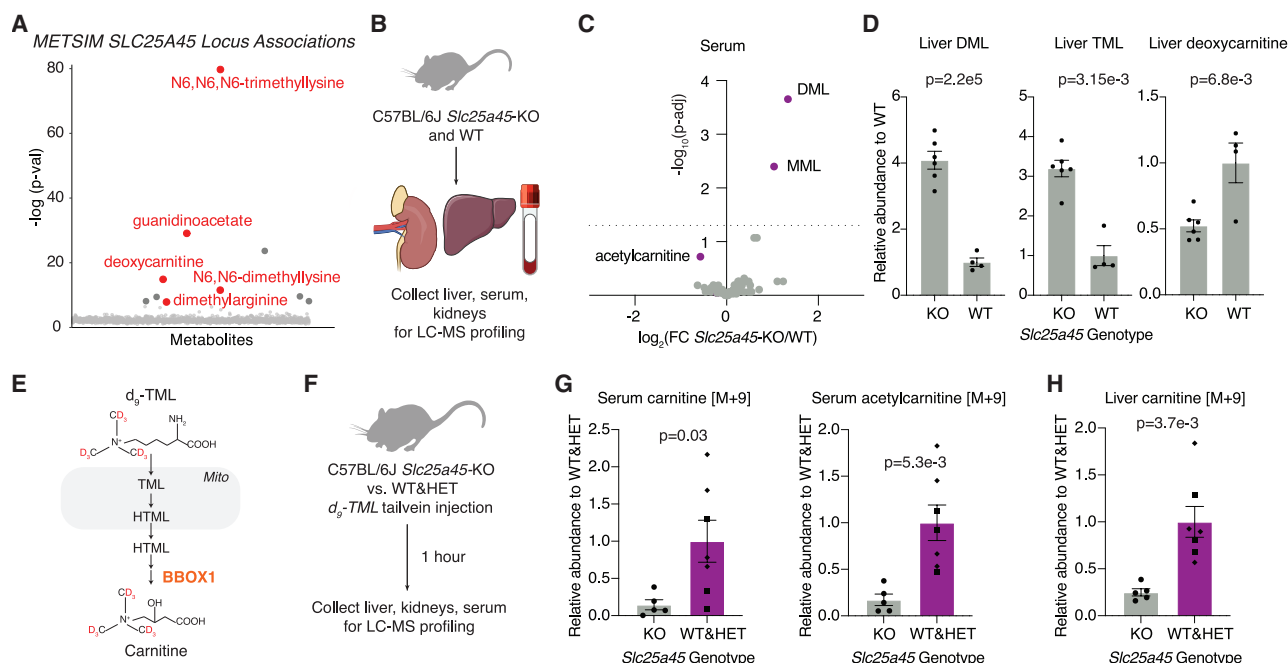


Figure 3. SLC25A45 is required for carnitine biosynthesis *in vivo*

(A) METSIM SLC25A45-metabolite associations displayed as $-\log(\min p \text{ value})$ for the given metabolite within the defined SLC25A45 locus vs. metabolites in alphabetical order.

(B) Schematic for the metabolite profiling of serum, kidney, and liver of *Slc25a45*-knockout and wild-type mice.

(C) Volcano plot with $-\log_{10}(p \text{ value})$ versus $\log_2(\text{fold change})$ in metabolite abundance normalized to glutamine levels in serum of *Slc25a45*-knockout ($n = 6$ biological replicates) and wild-type ($n = 4$ biological replicates) female mice. Statistical significance was determined by two-tailed unpaired *t* test followed by Benjamini-Hochberg multiple-test correction. The horizontal dotted line is the significance threshold of $p \text{ adj} < 0.05$.

(D) Abundances of DML, TML, and deoxycarnitine normalized to leucine in liver of *Slc25a45*-knockout ($n = 6$ biological replicates) and wild-type ($n = 4$ biological replicates) female mice relative to wild-type levels. Statistical significance was determined by two-tailed unpaired *t* test. Data are the mean $\pm$ SEM with the individual points shown.

(E) Schematic for the d_9 -TML isotope tracing.

(F) Schematic for the tail vein injection of isotope-labeled d_9 -TML tracer in the *Slc25a45*-knockout and control mice.

(G) Abundance of serum [M+9] carnitine and acetylcarnitine normalized to serum glutamine and [M+9] TML-labeled fraction in *Slc25a45*-knockout ($n = 5$, biological replicates, circle), heterozygous ($n = 3$ biological replicates, square), and wild-type ($n = 4$ biological replicates diamond) female mice relative to wild-type and heterozygous levels. Statistical significance was determined by two-tailed unpaired *t* test. Data are the mean $\pm$ SEM with the individual points shown.

(H) Abundance of liver [M+9] carnitine normalized to tryptophan and serum [M+9] TML-labeled fraction in *Slc25a45*-knockout ($n = 5$ biological replicates, circle), heterozygous ($n = 3$ biological replicates, square), and wild-type ($n = 4$, biological replicates, diamond) female mice relative to wild-type and heterozygous levels. Statistical significance was determined by two-tailed unpaired *t* test. Data are the mean $\pm$ SEM with the individual points shown.

body weight between genotypes (Figure S3C). Consistent with the GWAS data, we observed significantly altered levels of monomethyllysine (MML) and DML in the serum of *Slc25a45*-knockout mice. Whereas most of the carnitine biosynthesis pathway occurs in many organs, the terminal BBOX1-catalyzed reaction is predominantly restricted to liver and kidney.²² Thus, given the role of these organs in carnitine synthesis and high expression of *Slc25a45*²⁵ (Figure S3D), we examined how liver and kidney metabolism is affected by *Slc25a45* loss. Consistent with serum profiles, we observed significant changes in the levels of methylated amino acids. Additionally, deoxycarnitine levels, a downstream product of TML in the carnitine biosynthesis pathway, were reduced, supporting the findings from human GWAS serum data (Figures 3D and S4A). These results strongly suggest a role for SLC25A45 in carnitine biosynthesis *in vivo*.

To formally test this possibility, we performed d_9 -TML tracing into carnitine metabolites in *Slc25a45*-knockout and wild-type

mice via tail vein injection, followed by serum, liver, and kidney collection (Figures 3E and 3F). While we did not observe a difference in the labeled fraction of serum d_9 -TML (Figure S4B), the incorporation of deuterium atoms into carnitine species was decreased up to 35-fold in *Slc25a45*-knockout mice (Figures 3G, S4C, and S4D). Consistently, the [M+9] carnitine abundance was also reduced in the liver and kidneys harvested from *Slc25a45*-knockout mice (Figures 3H and S4E). Notably, the labeled fraction of d_9 -TML was elevated in the liver of *Slc25a45*-knockout mice (Figure S4F), possibly indicating that SLC25A45 regulates a major TML consumption pathway. Altogether, these findings show that SLC25A45 is required for carnitine biosynthesis and controls a critical route of TML utilization.

SLC25A45 supports carnitine synthesis under fasting conditions

During prolonged fasting and starvation, organisms rely on carnitine-dependent mitochondrial β -oxidation of fatty acids for

energy. Acetylcarnitine, the most abundant carnitine derivative, facilitates this process by shuttling acetyl groups across the mitochondrial membrane and buffering acetyl-CoA availability.²⁶ In line with this, d_9 -TML tracing in mice displayed a higher serum [M+9] acetylcarnitine/carnitine ratio under fasting conditions compared to *ad libitum* feeding in mice (Figures S4G and S4H). Given the requirement of SLC25A45 for carnitine synthesis, we hypothesized that its function may become critical during fasting, when carnitine intake is limited and demand is elevated. To test this, we subjected *Slc25a45*-knockout and wild-type mice to either a 24-h fast or *ad libitum* feeding, followed by serum and liver collection for polar and lipidomic profiling (Figure 4A). Consistent with the increased utilization of fatty acid oxidation upon starvation,²⁷ we observed a dramatic increase of serum 3-hydroxybutyrate (3-HB) levels upon fasting (Figure S4I). Additionally, the most altered serum metabolite between genotypes in response to fasting was acetylcarnitine (Figures S4J–S4M). *Slc25a45*-knockout mice failed to mount the fasting-induced increase in acetylcarnitine, indicative of potential carnitine limitation (Figure 4B). Consistently, livers from fasted *Slc25a45*-knockout mice showed reduced levels of carnitine and propionylcarnitine (Figures S4N–S4P). Lipidomic analysis further revealed significant reductions in several mid-to-long-chain acylcarnitines with additional species trending lower in the livers from *Slc25a45*-knockout mice (Figures 4C and S4Q). Given the well-established transcriptional response to fasting, such as the upregulation of β -oxidation and oxidative phosphorylation pathways,^{28,29} we next asked whether loss of *Slc25a45* impairs this adaptive program. We therefore used gene set enrichment analysis (GSEA) to identify β -oxidation and oxidative phosphorylation as signatures enriched in genes with differential fasting responses between livers from wild-type and *Slc25a45*-knockout mice. Targeted analysis of these pathways showed that *Slc25a45* loss attenuated the fasting-induced transcriptional upregulation of key genes involved in fatty acid oxidation and electron transport chain function (Figures 4D–4F). This raises the possibility that SLC25A45 is essential for the induction of transcriptional responses upon fasting. Consistent with this, the previous work reported liver-specific *Cpt1a*-knockout mice that functionally mimic carnitine deficiency also fail to mount acetyl-carnitine levels during fasting and show a blunted transcriptional fasting response.³⁰ Altogether, our data suggest a critical role for SLC25A45 in sustaining carnitine synthesis and fatty acid oxidation during fasting.

SLC25A45 mediates mitochondrial methylated amino acid import

To investigate the mechanism by which SLC25A45 impacts carnitine metabolism, we next examined its role in mitochondrial metabolism given its mitochondrial localization (Figure 5A). We crossed *Slc25a45*-knockout mice with the MITO-Tag model,³¹ which enables rapid immunopurification of mitochondria from tissues for multiomic analysis (Figure 5B). Using this system, we isolated mitochondria from liver and kidney for metabolite profiling and confirmed enrichment of mitochondrial proteins and metabolites with minimal contamination from other organelles (Figures S5A–S5D). LC-MS analysis revealed that mitochondria from *Slc25a45*^{−/−} mice exhibited markedly reduced levels of TML and ADMA along with a significant decrease in sac-

charopine, a key intermediate in lysine degradation (Figures 5C and 5D). These results suggest that SLC25A45 regulates mitochondrial levels of methylated basic amino acids.

To further confirm these results, we generated *SLC25A45*-knockout HeLa cells expressing a Mito-Tag construct (3×HA-OMP25-mCherry)³² (Figure S6A) and profiled the polar metabolome of immunopurified mitochondria by LC-MS. Consistent with *in vivo* experiments, *SLC25A45*-knockout cells showed a 9-fold depletion in mitochondrial TML levels and a 4-fold reduction in ADMA compared to controls (Figure 5E). Notably, *SLC25A45* loss did not substantially affect whole-cell levels of these metabolites (Figure S6B). We observed similar results in *SLC25A45*-knockout Jurkat cells as well as HEK293T and HeLa clonal knockouts complemented with *SLC25A45* cDNA (Figures 5F, S6A, and S6C–S6E). To directly assess the impact of *SLC25A45*-mediated TML import on carnitine production, we engineered HEK293T cells to express the carnitine biosynthetic enzyme BBOX1 (Figure S6F). Consistent with *in vivo* tracing experiments, *SLC25A45*-knockout cells exhibited a 3-fold decrease in the incorporation of d_9 -TML into carnitine (Figure S6G). These data raise the possibility that SLC25A45 may mediate mitochondrial import of methylated basic amino acids such as TML. To test this, we performed a mitochondrial uptake assay using isotope-labeled d_9 -TML and d_7 -ADMA in mitochondria isolated from control and *SLC25A45*-knockout cells (Figure 5G). Mitochondria immunopurified from *SLC25A45*-knockout cells exhibited a >50-fold reduction in uptake of both substrates compared to control cells (Figures 5H and 5I), confirming a role for SLC25A45 in the mitochondrial import of methylated basic amino acids. To further assess substrate specificity, we conducted competitive uptake experiments with labeled d_9 -TML in the presence of excess unlabeled substrates (Figure S6H). ADMA effectively competed with TML, followed by moderate and low competition from arginine and lysine, respectively (Figure S6I). Supporting arginine as a potential substrate, tracing with [1,2-¹³C₂]arginine revealed decreased levels of its downstream product, [M+2] ornithine, in mitochondria from *SLC25A45*-knockout cells (Figures S6J and S6K). These data are also consistent with depletion of hepatic guanidinoacetate, a mitochondrial intermediate of arginine metabolism and creatine biosynthesis,³³ in *Slc25a45*-knockout mice upon fasting (Figure S6L).

To gain better insight into the transport mechanism, we sought to identify the residues necessary for the import of methylated amino acids. We generated a homology model of SLC25A45 by aligning a predicted AlphaFold³⁴ model of SLC25A45 and the bovine ADP/ATP transporter³⁵ structure. This analysis pinpointed two potential substrate-binding residues, D121 and K124, which are conserved across multiple species (Figure 5J). While mutating these amino acids to alanine did not alter localization or expression (Figures S6E, S6M, and S6N), expression of mutant cDNAs failed to restore levels of mitochondrial TML (Figure 5K) and ADMA in *SLC25A45*-knockout cells, indicating the critical role of these residues in *SLC25A45*-mediated import (Figure 5L). Of note, we also tested the GWAS missense variant (rs34400381), which results in an R283C substitution in SLC25A45 and is associated with serum levels of methylated basic amino acids. Consistent with the GWAS findings, mutating R283 to cysteine, but not

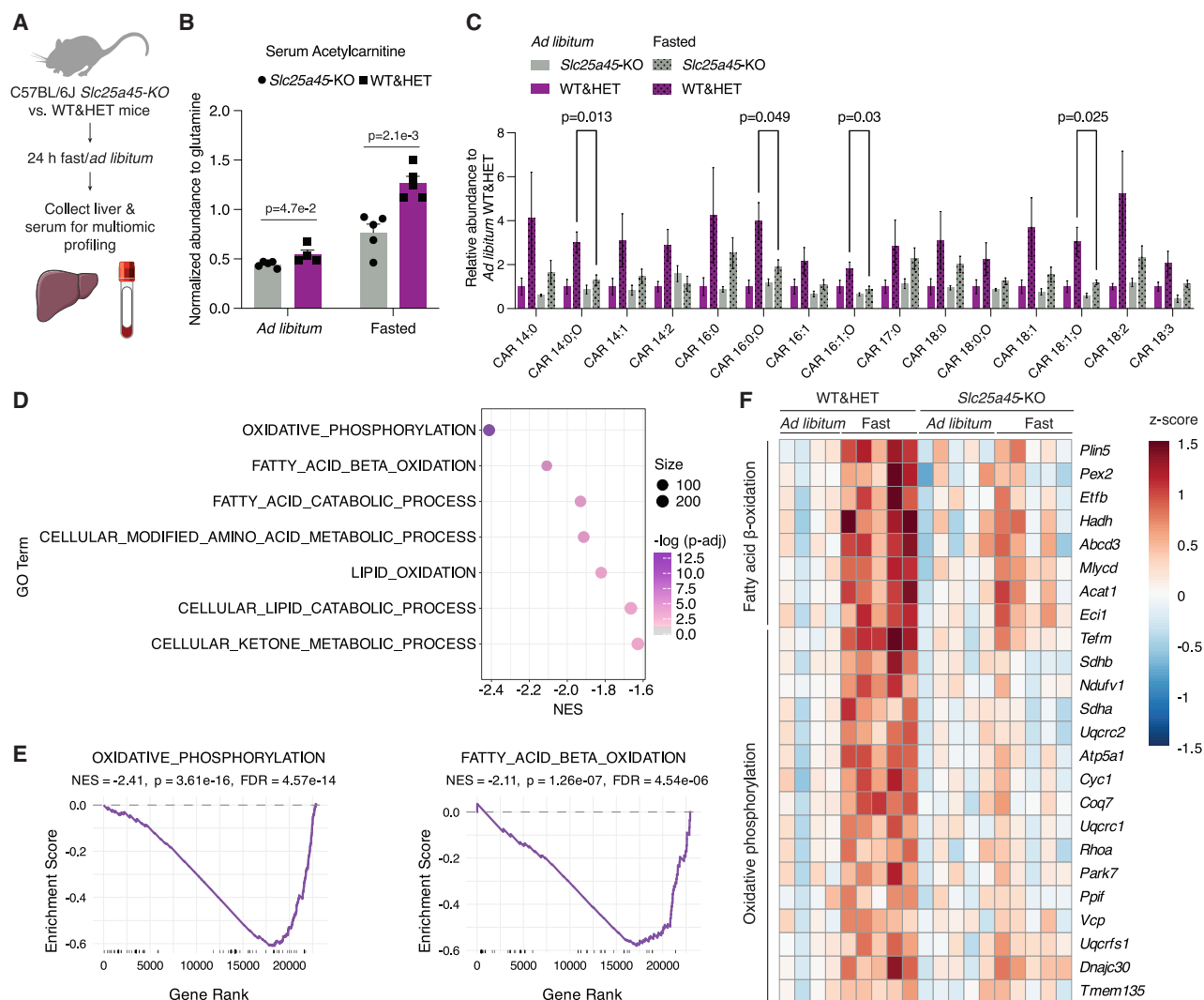


Figure 4. Metabolic response of *Slc25a45*-knockout mice upon fasting

(A) Schematic for the 24-h fasting experiment.

(B) The bar plot with the abundance of acetylcarnitine normalized to glutamine in the serum samples from *Slc25a45*-knockout ($n = 5$ ad libitum, $n = 5$ fasted biological replicates) vs. heterozygous ($n = 3$ ad libitum, $n = 3$ fasted biological replicates) and wild-type ($n = 1$ ad libitum, $n = 2$ fasted biological replicates) male mice upon fasting and ad libitum conditions. Data are the mean $\pm$ SEM with the individual points shown. Statistical significance was determined by two-tailed unpaired t test.

(C) The bar plots with the relative abundance of carnitine species ($>C10$ and $<C20$) normalized to wild-type and heterozygous ad libitum levels in the liver samples from *Slc25a45*-knockout ($n = 5$ ad libitum, $n = 5$ fasted biological replicates) vs. heterozygous ($n = 3$ ad libitum, $n = 3$ fasted biological replicates) and wild-type ($n = 1$ ad libitum, $n = 2$ fasted biological replicates) male mice upon fasting and ad libitum conditions. Data are the mean $\pm$ SEM with the individual points shown. Statistical significance was determined by the two-tailed unpaired t test.

(D) Gene set enrichment analysis (GSEA) of the transcriptional response in hepatic *Slc25a45*-knockout ($n = 5$ ad libitum, $n = 5$ fasted biological replicates) vs. heterozygous ($n = 3$ ad libitum, $n = 3$ fasted biological replicates) and wild-type ($n = 1$ ad libitum, $n = 2$ fasted biological replicates) male mice upon fasting and ad libitum conditions.

(E) Enrichment plots for the oxidative phosphorylation and fatty acid β -oxidation gene set enrichment terms and indicated statistics parameters.

(F) Heatmap with the indicated oxidative phosphorylation- and fatty acid β -oxidation-relevant transcripts in *Slc25a45*-knockout ($n = 5$ ad libitum, $n = 5$ fasted) vs. heterozygous ($n = 3$ ad libitum, $n = 3$ fasted biological replicates) and wild-type ($n = 1$ ad libitum, $n = 2$ fasted biological replicates) male mice upon fasting and ad libitum conditions. The values correspond to Z scores computed on the baseline normalized $\log_2$ transcript abundances.

alanine, altered mitochondrial TML levels, confirming the physiological relevance of this variant (Figure S6O). Collectively, our data strongly indicate that SLC25A45 mediates the mitochondrial import of methylated basic amino acid and supports carnitine synthesis.

Conclusion

The physiological substrates of many SLCs remain elusive, as traditional deorphanization approaches have relied on low-throughput candidate-based screening and comparative genomics. More recently, serum metabolite GWAS has enabled an

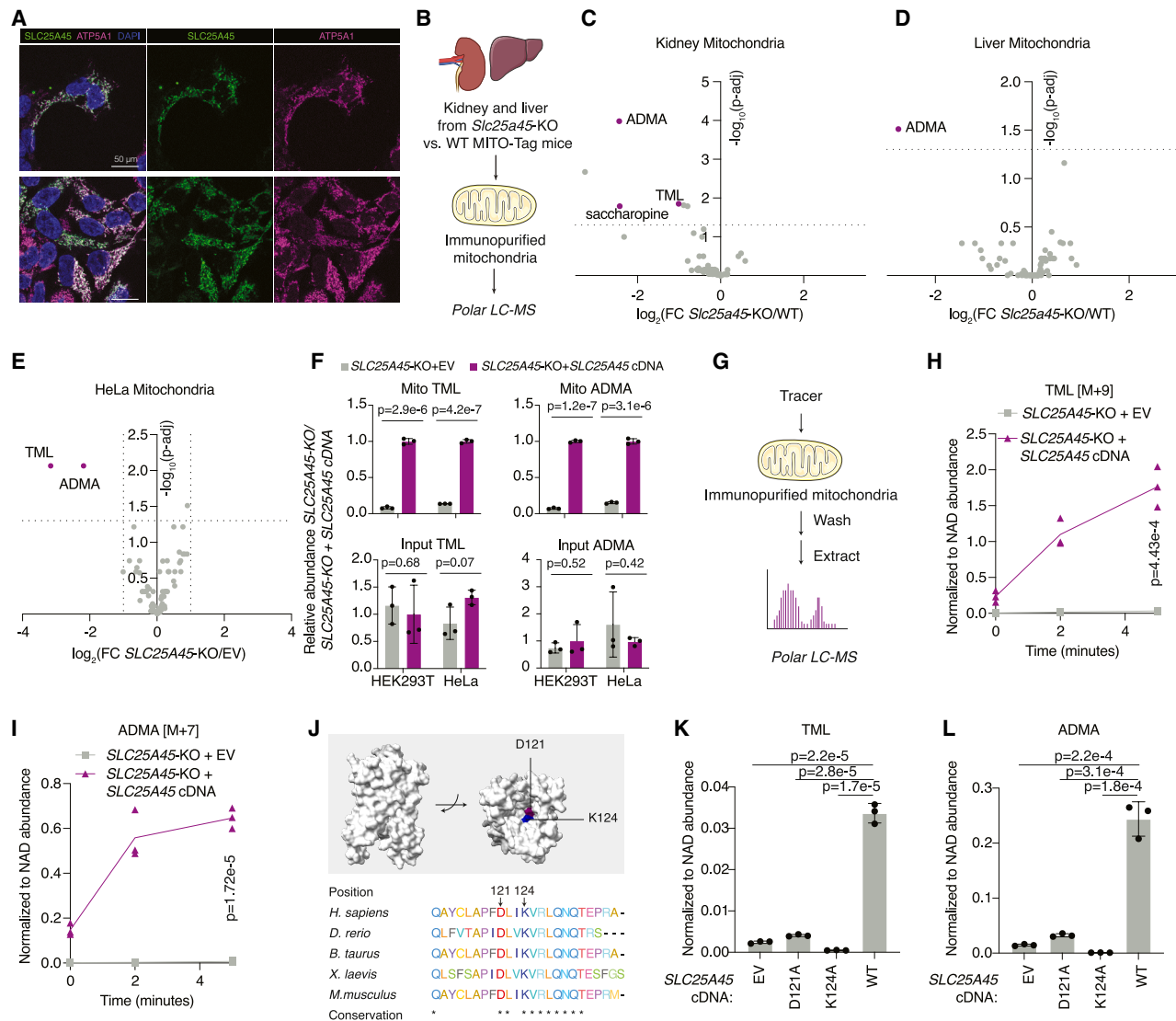


Figure 5. SLC25A45 mediates mitochondrial import of methylated amino acids

(A) Immunofluorescence analysis of ATP5A1 (magenta), SLC25A45 (green), and the nucleus (4,6-diamidino-2-phenylindole [DAPI] staining, blue) in HEK293T cells. Scale bar, 50 μ m.

(B) Schematic for the mitochondrial immunopurification from the kidney and liver of MITO-Tag *Slc25a45*-knockout and wild-type mice.

(C and D) Volcano plot with $-\log_{10}(p \text{ adj})$ versus $\log_2(\text{fold change})$ in metabolite abundance normalized to NAD levels in (C) kidney and (D) liver of *Slc25a45*-knockout ($n = 6$ biological replicates) and wild-type ($n = 4$ biological replicates) female mice. Statistical significance was determined by two-tailed unpaired t test followed by Benjamini-Hochberg multiple-test correction. The horizontal dotted line is the significance threshold of $p \text{ adj} < 0.05$.

(E) Volcano plot with $-\log_{10}(p \text{ adj})$ versus $\log_2(\text{fold change})$ in metabolite abundance normalized to NAD levels in immunopurified mitochondria from HeLa *SLC25A45*-knockout and corresponding control cells. The horizontal dotted line is the significance threshold of $p \text{ adj} < 0.05$. The vertical dotted line corresponds to $\text{abs}(\log_2\text{FC}) = 1$. Two-tailed unpaired t tests followed by Benjamini-Hochberg multiple-test correction were performed; $n = 3$ biological replicates.

(F) Relative metabolite abundance of input or mitochondrial ADMA and TML levels from *SLC25A45*-knockout HEK293T and HeLa cell lines expressing an empty vector control compared with those expressing *SLC25A45* cDNA. Data are the mean $\pm$ SD with the individual points shown. Statistical significance was determined by two-tailed unpaired t test; $n = 3$ biological replicates.

(G) Schematic showing the isotope tracer substrate uptake in the immunopurified mitochondria.

(H and I) d_9 -TML (H) and d_7 -ADMA (I) abundance in immunopurified mitochondria from HEK293T *SLC25A45*-knockout cells carrying an empty vector control (gray) or *SLC25A45* cDNA (purple) after incubation with 10 μ M tracer for the indicated time points. Data are individual points normalized to NAD abundance; $n = 3$ biological replicates. Lines correspond to the mean for each time point. Statistical significance was determined by two-tailed unpaired t test for the final time point.

(J) Top: AlphaFold model of SLC25A45 with the colored in purple residues (D121 and K124) for the following mutagenesis analysis. Bottom: sequence alignment of SLC25A45 homologs for the indicated residues. *D. rerio*, *Danio rerio*; *H. sapiens*, *Homo sapiens*; *M. musculus*, *Mus musculus*; *X. laevis*, *Xenopus laevis*; *B. taurus*, *Bovine taurus*. Asterisks indicate the conserved residues across the considered species.

(K and L) Normalized to NAD abundance of TML and ADMA in HEK293T *SLC25A45*-knockout cells expressing either an empty vector control or indicated *SLC25A45* cDNA. Data are the mean $\pm$ SD with the individual points shown. Statistical significance was determined by two-tailed unpaired t test. $n = 3$ biological replicates.

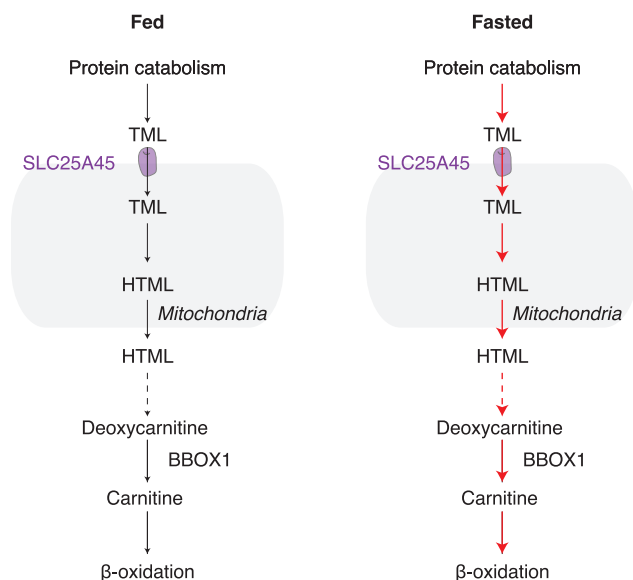


Figure 6. Proposed model for the role of SLC25A45 in carnitine biosynthesis

unbiased and high-throughput strategy to substrate discovery.^{3,4,36} A key advantage of this approach is its ability to detect subtle, physiologically relevant changes, including those masked by functional redundancy or tissue-specific expression. Leveraging GWAS datasets, we developed a supervised machine-learning guided prioritization framework that reduces dependence on stringent statistical filtering^{3,4} and integrates experimental metabolic pathway knowledge, rendering biologically interpretable predictions. For example, this approach prioritized an overlooked (by conventional GWASs) *SLC45A4*-acetyl-spermidine association that we experimentally validated as a candidate for polyamine transport. We also identified *UNC93A* as a mediator of plasma membrane acetylglucosamine transport. Given its high expression in skin, we speculate that *UNC93A* may contribute to the regulation of hyaluronic acid metabolism, a pathway essential for skin homeostasis.^{20,21} Likewise, we discovered *SLC25A45* as a mediator of methylated basic amino acid mitochondrial import, functionally annotating a critical transporter in the carnitine synthesis pathway.

Methylated amino acids have largely been regarded as inert byproducts of protein catabolism, with little consideration for their metabolic significance. Although carnitine synthesis from isotope-labeled TML was described in 1961,³⁷ the molecular players underlying mitochondrial import of TML, the first step in this pathway, have remained elusive. Carnitine is primarily obtained through diet, but during nutrient deprivation and starvation, endogenous carnitine synthesis becomes critical, particularly to sustain β -oxidation. Consistent with this, *SLC25A45* deficiency partially phenocopies the blunted hepatic acetylcarnitine accumulation as well as the transcriptional response seen in liver-specific *Cpt1*-knockout mice.³⁰ Building upon this, we speculate that *SLC25A45* confers an evolutionary advantage during starvation by channeling protein breakdown products into β -oxidation via carnitine synthesis (Figure 6). Future work needs to establish the potential regulatory mechanisms that co-

ordinate protein catabolism, carnitine biosynthesis, and β -oxidation in response to such stress conditions. This will also provide further insight into how organisms and metabolic pathways have evolved to efficiently utilize resources through the recycling of metabolic products. Beyond fasting, *SLC25A45*-dependent carnitine biosynthesis may be critical under dietary regimens such as vegetarian ketogenic diets, where intake of carnitine is limited and demand is increased. This is evident from the lethality observed in liver-specific *Cpt2*-knockout mice, which fail to survive beyond 1 week on a ketogenic diet.³⁸ While we focused on hepatic β -oxidation, mitochondrial fatty acid oxidation is also critical in other energy-demanding tissues. For example, skeletal muscle-specific *Cpt1b* knockout mice exhibit compensatory peroxisomal oxidation.³⁹ This observation warrants further investigation of the function of *SLC25A45* in carnitine biosynthesis and β -oxidation across other tissues.

Although we primarily investigated TML, our findings strongly indicate that *SLC25A45* has other physiologically relevant substrates. For example, further studies should examine the role of ADMA and its implications for mitochondrial nitric oxide synthesis and cardiovascular health.²³ Our transport assays suggest that arginine is also likely a substrate of *SLC25A45*. Indeed, upon fasting of *Slc25a45*-knockout mice, we observed a larger depletion of guanidinoacetate, a downstream product of mitochondrial arginine and an intermediate in creatine biosynthesis. This raises intriguing questions about potential synergistic interactions between creatine and carnitine pathways within mitochondria that could also contribute to the observed fasting response. Moreover, *SLC25A45* may also transport methylated forms of basic amino acids, which can serve as mitochondrial sources of lysine. Multiple biochemical studies reported the existence of lysine and arginine demethylases in mitochondria, although their identity to date remains elusive.^{40,41} Consistent with this, despite significant changes in lysine downstream metabolites, such as saccharopine, lysine itself was not identified as a substrate for *SLC25A45*. Thus, future structural studies should elucidate the mechanistic basis of *SLC25A45* function and its substrate specificity.

Finally, it is critical to distinguish the role of *SLC25A45* from other transporters with overlapping substrates, such as a mitochondrial basic amino acid importer *SLC25A29*. While arginine is a substrate of both carriers, *SLC25A45* exhibits higher expression than *SLC25A29* in mouse liver and kidney.⁴² Such differential tissue expression pattern suggests that distinct physiological contexts may require tailored transport activities, driven by variations in substrate affinity, specificity, or even regulatory mechanisms. Understanding how these transporters coordinate mitochondrial amino acid metabolism will be key to uncovering new metabolic vulnerabilities and compartmentalization mechanisms.

Limitations of the study

Our supervised machine-learning model is trained on previously reported gene-metabolite pairs, which may not fully capture the broader biological landscape and could limit generalizability. Despite the safeguards, residual data leakage and overfitting cannot be completely ruled out. Moreover, our GWAS-derived features come exclusively from European-ancestry cohorts, underscoring the need to validate and extend the pipeline in other populations to ensure ancestry-agnostic performance. Further

work is necessary to benchmark the proposed machine-learning approach against alternative methods for GWAS analysis.

Metabolite tracing experiments strongly suggest the transport function of the candidate cellular and mitochondrial SLCs identified in the study. However, further structural studies as well as liposome-based *in vitro* reconstituted uptake for SLC25A45 and *Xenopus laevis* oocytes assays for SLC45A4 and UNC93A should be performed to definitively study the function of these transporters. Additionally, *Slc25a45*-knockout mice in this study were generated on the C57BL/6J background, containing a well-characterized mutation in nicotinamide nucleotide transhydrogenase (NNT) that may affect mitochondrial phenotypes.⁴³ Further characterizations of *Slc25a45*-knockout mice should consider this aspect.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Kıvanç Birsoy (kbirsoy@rockefeller.edu).

Materials availability

All unique reagents generated in this study are available from the lead contact upon request.

Data and code availability

Uncropped western blots and all data used to generate display items in this manuscript are available in [Data S1](#). Accession numbers are listed in the [key resources table](#). RNA-seq data have been deposited at Gene Expression Omnibus (GEO: GEO 308565) and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#). LC-MS files have been uploaded to Metabolomics Workbench⁴⁴ (PR002544). Computational pipeline files are available on GitHub (https://github.com/artemkhan/SLC_ML) and Zenodo (<https://doi.org/10.5281/zenodo.16067534>). Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

ACKNOWLEDGMENTS

We thank all members of the Birsoy Lab for suggestions and also members of the Rockefeller University Proteomics Resource Center and the Flow Cytometry Resource Center. We thank Brandon Chen, Alexander Tarakhovskiy, Adam Wang, and Zihan Xu for the helpful discussion and feedback. We thank Chingwen Yang and Chia-Yun Han from the Gene Targeting Resource Center for the generation of *Slc25a45*-knockout mice. We thank summer undergraduate student Dio Sangwon Shin for help with SLC45A4. We thank Rada Norinsky from the Rockefeller University Transgenic Services department for performing *in vitro* fertilization to expand the mouse colony. Some figures use modified illustrations from Servier Medical Art and publicly available NIH BIOART Source licensed under a Creative Commons Attribution 3.0 Unported License. A.K. is supported by a Boehringer Ingelheim Fonds PhD Fellowship and National Institute Of Diabetes And Digestive And Kidney Diseases of the National Institutes of Health under award number F99DK144923. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. F.S.Y. and M.X. are supported by a Medical Scientist Training Program grant from the National Institute of General Medical Sciences of the National Institutes of Health under award number T32GM152349 to the Weill Cornell/Rockefeller/Sloan Kettering Tri-Institutional MD-PhD Program. G.U. is a Damon Runyon Fellow supported by the Damon Runyon Cancer Research Foundation (DRG-2431-21) and NIH/NCI 1K99CA286722-01. E.R.G. is supported by grants from the NIH (R01HG011138 and R01GM140287), a Vanderbilt Diabetes Center Functional Genomics Core grant, and a gift from the Scott Hamilton CARES Foundation (in honor of Carlee Vaughn). K.B. is supported by NIDDK (R01 DK140337),

G. Harold and Leila Y. Mathers Foundation, and Mark Foundation Emerging Leader Award and is a Pew Innovation Fund Investigator.

AUTHOR CONTRIBUTIONS

K.B., A.K., F.S.Y., and G.U. conceived the project. K.B. and A.K. wrote the manuscript. K.B. and A.K. designed experiments. A.K. developed the machine-learning platform with input from E.R.G. A.K. performed UNC93A and SLC45A4 experiments. A.K., F.S.Y., G.U., N.L.D., R.E., M.X., and K.W. performed SLC25A45 experiments. K.C. and G.J.P. conducted lipidomic analysis.

DECLARATION OF INTERESTS

K.B. is scientific advisor to Nanocare Pharmaceuticals and Atavistik Bio. E.R.G. has consulted for Thryv Therapeutics and is a co-inventor on patents for molecular signatures of cardiovascular phenotypes and metabolic health and the use of RNAs as therapeutics and diagnostic biomarkers. G.J.P. and K.C. have a collaborative research agreement with Thermo Fisher Scientific, and G.J.P. is a scientific advisory board member for Cambridge Isotope Laboratories and the chief scientific officer of Panome Bio.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT in order to edit code and manuscript text. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- [KEY RESOURCES TABLE](#)
- [EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS](#)
 - Animals
- [METHOD DETAILS](#)
 - Generation of the machine learning prediction pipeline
 - Cell lines and reagents
 - Generation of knockout and cDNA-overexpression cell lines
 - Immunofluorescence staining and confocal microscopy for co-localization
 - Immunoblotting
 - Whole cell metabolomics experiments
 - Polar metabolite LC-MS profiling
 - LC/MS analysis for lipidomics
 - Immunopurified mitochondria metabolic profiling from the cells and mouse tissues
 - Isotope labeled uptake in immunopurified mitochondria
 - Mouse work
 - Mouse serum for metabolomic analysis
 - Tissues for polar metabolomic and lipidomic analysis
 - *In vivo* d₉-TML tracing
 - Fasting experiments
 - RNA-sequencing
- [QUANTIFICATION AND STATISTICAL ANALYSIS](#)
 - Statistical analysis and data visualization

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cmet.2025.09.015>.

Received: April 2, 2025

Revised: September 25, 2025

Accepted: September 30, 2025

Published: October 10, 2025

REFERENCES

- Khan, A., Liu, Y., Gad, M., Kenny, T.C., and Birsoy, K. (2025). Solute carriers: The gatekeepers of metabolism. *Cell* 188, 869–884. <https://doi.org/10.1016/j.cell.2025.01.015>.
- César-Razquin, A., Snijder, B., Frappier-Brinton, T., Isserlin, R., Gyimesi, G., Bai, X., Reithmeier, R.A., Hepworth, D., Hediger, M.A., Edwards, A. M., and Superti-Furga, G. (2015). A call for systematic research on solute carriers. *Cell* 162, 478–487. <https://doi.org/10.1016/j.cell.2015.07.022>.
- Kenny, T.C., Khan, A., Son, Y., Yue, L., Heissel, S., Sharma, A., Pasolli, H. A., Liu, Y., Gamazon, E.R., Alwaseem, H., et al. (2023). Integrative genetic analysis identifies FLVCR1 as a plasma-membrane choline transporter in mammals. *Cell Metab.* 35, 1057–1071.e12. <https://doi.org/10.1016/j.cmet.2023.04.003>.
- Khan, A., Unlu, G., Lin, P., Liu, Y., Kilic, E., Kenny, T.C., Birsoy, K., and Gamazon, E.R. (2024). Metabolic gene function discovery platform GeneMAP identifies SLC25A48 as necessary for mitochondrial choline import. *Nat. Genet.* 56, 1614–1623. <https://doi.org/10.1038/s41588-024-01827-2>.
- Rebouche, C.J., and Engel, A.G. (1980). Tissue distribution of carnitine biosynthetic enzymes in man. *Biochim. Biophys. Acta* 630, 22–29. [https://doi.org/10.1016/0304-4165\(80\)90133-6](https://doi.org/10.1016/0304-4165(80)90133-6).
- van Vlies, N., Ofman, R., Wanders, R.J.A., and Vaz, F.M. (2007). Submitochondrial localization of 6-N-trimethyllysine dioxygenase – implications for carnitine biosynthesis. *FEBS J.* 274, 5845–5851. <https://doi.org/10.1111/j.1742-4658.2007.06108.x>.
- Zhou, D., Jiang, Y., Zhong, X., Cox, N.J., Liu, C., and Gamazon, E.R. (2020). A unified framework for joint-tissue transcriptome-wide association and Mendelian randomization analysis. *Nat. Genet.* 52, 1239–1246. <https://doi.org/10.1038/s41588-020-0706-2>.
- Yin, X., Chan, L.S., Bose, D., Jackson, A.U., VandeHaar, P., Locke, A.E., Fuchsberger, C., Stringham, H.M., Welch, R., Yu, K., et al. (2022). Genome-wide association studies of metabolites in Finnish men identify disease-relevant loci. *Nat. Commun.* 13, 1644. <https://doi.org/10.1038/s41467-022-29143-5>.
- Chen, Y., Lu, T., Pettersson-Kymmer, U., Stewart, I.D., Butler-Laporte, G., Nakanishi, T., Cerani, A., Liang, K.Y.H., Yoshiji, S., Willett, J.D.S., et al. (2023). Genomic atlas of the plasma metabolome prioritizes metabolites implicated in human diseases. *Nat. Genet.* 55, 44–53. <https://doi.org/10.1038/s41588-022-01270-1>.
- Rentzsch, P., Witten, D., Cooper, G.M., Shendure, J., and Kircher, M. (2019). CADD: predicting the deleteriousness of variants throughout the human genome. *Nucleic Acids Res.* 47, D886–D894. <https://doi.org/10.1093/nar/gky1016>.
- Torrents, D., Estévez, R., Pineda, M., Fernández, E., Lloberas, J., Shi, Y.B., Zorzano, A., and Palacín, M. (1998). Identification and characterization of a membrane protein (y+L amino acid transporter-1) that associates with 4F2hc to encode the amino acid transport activity y+L. A candidate gene for lysinuric protein intolerance. *J. Biol. Chem.* 273, 32437–32445. <https://doi.org/10.1074/jbc.273.49.32437>.
- Bröer, A., Wagner, C.A., Lang, F., and Bröer, S. (2000). The heterodimeric amino acid transporter 4F2hc/y+LAT2 mediates arginine efflux in exchange with glutamine. *Biochem. J.* 349, 787–795. Pt 3. <https://doi.org/10.1042/bj3490787>.
- Chubb, S., Kingsland, A.L., Bröer, A., and Bröer, S. (2006). Mutation of the 4F2 heavy-chain carboxy terminus causes y+ LAT2 light-chain dysfunction. *Mol. Membr. Biol.* 23, 255–267. <https://doi.org/10.1080/09687860600652968>.
- Chua, G.-L., Tan, B.C., Loke, R.Y.J., He, M., Chin, C.-F., Wong, B.H., Kuk, A.C.Y., Ding, M., Wenk, M.R., Guan, L., et al. (2023). Mfsd2a utilizes a flipase mechanism to mediate omega-3 fatty acid lysolipid transport. *Proc. Natl. Acad. Sci. USA* 120, e2215290120. <https://doi.org/10.1073/pnas.2215290120>.
- Gahl, W.A., and Pitot, H.C. (1979). Putrescine-oxidase activity in adult bovine serum and fetal bovine serum. *In Vitro* 15, 252–257. <https://doi.org/10.1007/BF02618948>.
- Middleton, S.J., Markússon, S., Åkerlund, M., Deme, J.C., Tseng, M., Li, W., Zuberi, S.R., Kuteyi, G., Sarkies, P., Baskozos, G., et al. (2025). SLC45A4 is a pain gene encoding a neuronal polyamine transporter. *Nature*. Published online August 20, 2025. <https://doi.org/10.1038/s41586-025-09326-y>.
- Wiedmer, T., Teoh, S.T., Christodoulaki, E., Wolf, G., Tian, C., Sedlyarov, V., Jarret, A., Leippe, P., Frommelt, F., Ingles-Prieto, A., et al. (2025). Metabolic mapping of the human solute carrier superfamily. *Mol. Syst. Biol.* 21, 560–598. <https://doi.org/10.1038/s44320-025-00106-4>.
- Love, D.C., and Hanover, J.A. (2005). The hexosamine signaling pathway: deciphering the “O-GlcNAc code.”. *Sci. STKE* 2005, re13. <https://doi.org/10.1126/stke.3122005re13>.
- Varki, A., Cummings, R., Esko, J., Freeze, H., Hart, G., and Marth, J. (1999). Degradation and turnover of glycans. In *Essentials of Glycobiology* (Cold Spring Harbor Laboratory Press).
- Ogston, A.G. (1964). On the estimation of glucosamine in hyaluronic acid. *Anal. Biochem.* 8, 337–343. [https://doi.org/10.1016/0003-2697\(64\)90066-1](https://doi.org/10.1016/0003-2697(64)90066-1).
- Price, R.D., Myers, S., Leigh, I.M., and Navsaria, H.A. (2005). The role of hyaluronic acid in wound healing: assessment of clinical evidence. *Am. J. Clin. Dermatol.* 6, 393–402. <https://doi.org/10.2165/00128071-200506060-00006>.
- Uhlén, M., Fagerberg, L., Hallström, B.M., Lindskog, C., Oksvold, P., Mardinoglu, A., Sivertsson, Å., Kampf, C., Sjöstedt, E., Asplund, A., et al. (2015). Tissue-based map of the human proteome. *Science* 347, 1260419. <https://doi.org/10.1126/science.1260419>.
- Böger, R.H. (2004). Asymmetric dimethylarginine, an endogenous inhibitor of nitric oxide synthase, explains the “L-arginine paradox” and acts as a novel cardiovascular risk factor. *J. Nutr.* 134, 2842S–2847S. <https://doi.org/10.1093/jn/134.10.2842S>.
- Vaz, F.M., and Wanders, R.J.A. (2002). Carnitine biosynthesis in mammals. *Biochem. J.* 361, 417–429. <https://doi.org/10.1042/0264-6021:3610417>.
- Cardoso-Moreira, M., Halbert, J., Valloton, D., Velten, B., Chen, C., Shao, Y., Liechti, A., Ascensão, K., Rummel, C., Ovchinnikova, S., et al. (2019). Gene expression across mammalian organ development. *Nature* 571, 505–509. <https://doi.org/10.1038/s41586-019-1338-5>.
- Stephens, F.B., Constantin-Teodosiu, D., and Greenhaff, P.L. (2007). New insights concerning the role of carnitine in the regulation of fuel metabolism in skeletal muscle. *J. Physiol.* 581, 431–444. <https://doi.org/10.1113/jphysiol.2006.125799>.
- Newman, J.C., and Verdin, E. (2014). β -hydroxybutyrate: Much more than a metabolite. *Diabetes Res. Clin. Pract.* 106, 173–181. <https://doi.org/10.1016/j.diabres.2014.08.009>.
- de la Calle Arregui, C., Plata-Gómez, A.B., Deleyto-Seldas, N., García, F., Ortega-Molina, A., Abril-Garrido, J., Rodríguez, E., Nemazanyy, I., Tribouillard, L., de Martino, A., et al. (2021). Limited survival and impaired hepatic fasting metabolism in mice with constitutive Rag GTPase signaling. *Nat. Commun.* 12, 3660. <https://doi.org/10.1038/s41467-021-23857-8>.
- Liu, L., Nam, M., Fan, W., Akie, T.E., Hoaglin, D.C., Gao, G., Keaney, J.F., Jr., and Cooper, M.P. (2014). Nutrient sensing by the mitochondrial transcription machinery dictates oxidative phosphorylation. *J. Clin. Investig.* 124, 768–784. <https://doi.org/10.1172/JCI69413>.
- Choi, J., Smith, D.M., Scafidi, S., Riddle, R.C., and Wolfgang, M.J. (2024). Carnitine palmitoyltransferase 1 facilitates fatty acid oxidation in a non-cell-autonomous manner. *Cell Rep.* 43, 115006. <https://doi.org/10.1016/j.celrep.2024.115006>.
- Bayraktar, E.C., Baudrier, L., Özerdem, C., Lewis, C.A., Chan, S.H., Kunchok, T., Abu-Remaih, M., Cangelosi, A.L., Sabatini, D.M., Birsoy, K., and Chen, W.W. (2019). MITO-Tag Mice enable rapid isolation and

- multimodal profiling of mitochondria from specific cell types in vivo. *Proc. Natl. Acad. Sci. USA* 116, 303–312. <https://doi.org/10.1073/pnas.1816656115>.
32. Chen, W.W., Freinkman, E., and Sabatini, D.M. (2017). Rapid immunopurification of mitochondria for metabolite profiling and absolute quantification of matrix metabolites. *Nat. Protoc.* 12, 2215–2231. <https://doi.org/10.1038/nprot.2017.104>.
33. da Silva, R.P., Nissim, I., Brosnan, M.E., and Brosnan, J.T. (2009). Creatine synthesis: hepatic metabolism of guanidinoacetate and creatine in the rat in vitro and in vivo. *Am. J. Physiol. Endocrinol. Metab.* 296, E256–E261. <https://doi.org/10.1152/ajpendo.90547.2008>.
34. Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., Tunyasuvunakool, K., Bates, R., Židek, A., Potapenko, A., et al. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature* 596, 583–589. <https://doi.org/10.1038/s41586-021-03819-2>.
35. Pebay-Peyroula, E., Dahout-Gonzalez, C., Kahn, R., Trézéguet, V., Lauquin, G.J.-M., and Brandolin, G. (2003). Structure of mitochondrial ADP/ATP carrier in complex with carboxyatractyloside. *Nature* 426, 39–44. <https://doi.org/10.1038/nature02056>.
36. Suhre, K., Shin, S.-Y., Petersen, A.-K., Mohney, R.P., Meredith, D., Wägele, B., Altmaier, E., CARDIoGRAM, Deloukas, P., Erdmann, J., et al. (2011). Human metabolic individuality in biomedical and pharmaceutical research. *Nature* 477, 54–60. <https://doi.org/10.1038/nature10354>.
37. Wolf, G., and Berger, C.R. (1961). Studies on the biosynthesis and turnover of carnitine. *Arch. Biochem. Biophys.* 92, 360–365. [https://doi.org/10.1016/0003-9861\(61\)90362-9](https://doi.org/10.1016/0003-9861(61)90362-9).
38. Lee, J., Choi, J., Scafidi, S., and Wolfgang, M.J. (2016). Hepatic fatty acid oxidation restrains systemic catabolism during starvation. *Cell Rep.* 16, 201–212. <https://doi.org/10.1016/j.celrep.2016.05.062>.
39. Wicks, S.E., Vandanmagsar, B., Haynie, K.R., Fuller, S.E., Warfel, J.D., Stephens, J.M., Wang, M., Han, X., Zhang, J., Noland, R.C., and Mynatt, R.L. (2015). Impaired mitochondrial fat oxidation induces adaptive remodeling of muscle metabolism. *Proc. Natl. Acad. Sci. USA* 112, E3300–E3309. <https://doi.org/10.1073/pnas.1418560112>.
40. Paik, W.K., and Kim, S. (1973). Enzymatic demethylation of calf thymus histones. *Biochem. Biophys. Res. Commun.* 51, 781–788. [https://doi.org/10.1016/0006-291X\(73\)91383-1](https://doi.org/10.1016/0006-291X(73)91383-1).
41. Paik, W.K., and Kim, S. (1974). Epsilon-alkyllysine. New assay method, purification, and biological significance. *Arch. Biochem. Biophys.* 165, 369–378. [https://doi.org/10.1016/0003-9861\(74\)90175-1](https://doi.org/10.1016/0003-9861(74)90175-1).
42. Porcelli, V., Fiermonte, G., Longo, A., and Palmieri, F. (2014). The human gene SLC25A29, of solute carrier family 25, encodes a mitochondrial transporter of basic amino acids. *J. Biol. Chem.* 289, 13374–13384. <https://doi.org/10.1074/jbc.M114.547448>.
43. Ronchi, J.A., Figueira, T.R., Ravagnani, F.G., Oliveira, H.C.F., Vercesi, A. E., and Castilho, R.F. (2013). A spontaneous mutation in the nicotinamide nucleotide transhydrogenase gene of C57BL/6J mice results in mitochondrial redox abnormalities. *Free Radic. Biol. Med.* 63, 446–456. <https://doi.org/10.1016/j.freeradbiomed.2013.05.049>.
44. Sud, M., Fahy, E., Cotter, D., Azam, K., Vadivelu, I., Burant, C., Edison, A., Fiehn, O., Higashi, R., Nair, K.S., et al. (2016). Metabolomics Workbench: An international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools. *Nucleic Acids Res.* 44, D463–D470. <https://doi.org/10.1093/nar/gkv1042>.
45. Sergushichev, A.A. (2016). An algorithm for fast preranked gene set enrichment analysis using cumulative statistic calculation. Preprint at bioRxiv. <https://doi.org/10.1101/060012>.
46. Karp, P.D., Billington, R., Caspi, R., Fulcher, C.A., Latendresse, M., Kothari, A., Keseler, I.M., Krummenacker, M., Midford, P.E., Ong, Q., et al. (2019). The BioCyc collection of microbial genomes and metabolic pathways. *Brief. Bioinform.* 20, 1085–1093. <https://doi.org/10.1093/bib/bbx085>.
47. TensorFlow Developers (2025). TensorFlow (Zenodo). <https://doi.org/10.5281/zenodo.15009305> Version v2.19.0 .
48. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. (2011). Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.* 12, 2825–2830.
49. Smith, C.A., Want, E.J., O’Maille, G., Abagyan, R., and Siuzdak, G. (2006). XCMS: processing mass spectrometry data for metabolite profiling using nonlinear peak alignment, matching, and identification. *Anal. Chem.* 78, 779–787. <https://doi.org/10.1021/ac051437y>.
50. Adams, K.J., Pratt, B., Bose, N., Dubois, L.G., St. John-Williams, L., Perrott, K.M., Ky, K., Kapahi, P., Sharma, V., MacCoss, M.J., et al. (2020). Skyline for small molecules: a unifying software package for quantitative metabolomics. *J. Proteome Res.* 19, 1447–1458. <https://doi.org/10.1021/acs.jproteome.9b00640>.
51. S.R. Singh, R.M. Hoffman, and A. Singh, eds. (2021). *Mouse Genetics: Methods and Protocols* (Springer).
52. Timson, R.C., Khan, A., Uygur, B., Saad, M., Yeh, H.-W., DelGaudio, N.L., Weber, R., Alwaseem, H., Gao, J., Yang, C., and Birsoy, K. (2024). Development of a mouse model expressing a bifunctional glutathione-synthesizing enzyme to study glutathione limitation in vivo. *J. Biol. Chem.* 300, 105645. <https://doi.org/10.1016/j.jbc.2024.105645>.
53. Wang, W., Carroll, T. (2025). Rfastp Bioconductor. <https://doi.org/10.18129/B9.bioc.Rfastp>.
54. Patro, R., Duggal, G., Love, M.I., Irizarry, R.A., and Kingsford, C. (2017). Salmon provides fast and bias-aware quantification of transcript expression. *Nat. Methods* 14, 417–419. <https://doi.org/10.1038/nmeth.4197>.
55. Love, M.I., Huber, W., and Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol.* 15, 550. <https://doi.org/10.1186/s13059-014-0550-8>.
56. Gene Ontology Consortium, Aleksander, S.A., Balhoff, J., Carbon, S., Cherry, J.M., Drabkin, H.J., Ebert, D., Feuermann, M., Gaudet, P., Harris, N.L., et al. (2023). The Gene Ontology knowledgebase in 2023. *Genetics* 224, iyad031. <https://doi.org/10.1093/genetics/iyad031>.
57. Ashburner, M., Ball, C.A., Blake, J.A., Botstein, D., Butler, H., Cherry, J.M., Davis, A.P., Dolinski, K., Dwight, S.S., Eppig, J.T., et al. (2000). Gene Ontology: tool for the unification of biology. *Nat. Genet.* 25, 25–29. <https://doi.org/10.1038/75556>.
58. Meng, E.C., Goddard, T.D., Pettersen, E.F., Couch, G.S., Pearson, Z.J., Morris, J.H., and Ferrin, T.E. (2023). UCSF ChimeraX: Tools for structure building and analysis. *Protein Sci.* 32, e4792. <https://doi.org/10.1002/pro.4792>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit anti-HA-Tag	Cell Signaling Technology	Cat# 3724S; RRID: AB_1549585
Rabbit anti-Citrate Synthase	Cell Signaling Technology	Cat# 14309S; RRID: AB_2798421
Rabbit anti-SLC25A12	Abcam	Cat# ab200201
Mouse anti-GAPDH	GeneTex	Cat# GTX627408
Mouse anti-Flag-Tag	Sigma-Aldrich	Cat# F1804; RRID: AB_262044
Rabbit anti-SLC25A45	Invitrogen	Cat# PA5-62251; RRID: AB_2647421
Mouse anti-ATP5A1	Thermo Fisher Scientific	Cat# 439800; RRID: AB_2533548
Rat anti-LAMP-1 (clone 1D4B)	DSHB (University of Iowa)	Cat# 1D4B; RRID: AB_11149501
Rabbit anti-Calreticulin	GeneTex	Cat# GTX83089
HRP-conjugated anti-mouse IgG	Cell Signaling Technology	Cat# 7076S; RRID: AB_330924
HRP-conjugated anti-rabbit IgG	Cell Signaling Technology	Cat# 7074S; RRID: AB_2099233
HRP-conjugated anti-rat IgG	Cell Signaling Technology	Cat# 7077S; RRID: AB_10694715
Goat anti-mouse Alexa Fluor 488	Thermo Fisher Scientific	Cat# A11029; RRID: AB_2534088
Goat anti-rabbit Alexa Fluor 555	Thermo Fisher Scientific	Cat# A21428; RRID: AB_2535849
Donkey anti-rabbit Alexa Fluor 488	Thermo Fisher Scientific	Cat# A21206; RRID: AB_2535792
Donkey anti-mouse Alexa Fluor 555	Thermo Fisher Scientific	Cat# A10037; RRID: AB_2534013
Chemicals, peptides, and recombinant proteins		
Anti-HA magnetic beads	Thermo Scientific (Pierce)	Cat# 88837
DAPI	Thermo Fisher Scientific	Cat# D1306
Polybrene	Sigma-Aldrich	Cat# H9268
Puromycin	Sigma-Aldrich	Cat# P8833
Blasticidin	InvivoGen	Cat# ant-bl-1
Phalloidin-iFluor 647	ConjugateCayman Chemical Company	Cat# 20555
d ₉ -TML	Toronto Chemical	Cat# T796447
T4 DNA Ligase	New England Biolabs	Cat# M0202
BamHI-HF	New England Biolabs	Cat# R3136L
EcoRI-HF	New England Biolabs	Cat# R3101L
Gibson Assembly Master Mix	New England Biolabs	Cat# E2611
X-tremeGENE 9 DNA Transfection Reagent	Roche	Cat# 636478700
2-[¹⁵ N]amino-2-deoxy-D-[UL- ¹³ C ₆]glucose hydrochloride (D-[UL- ¹³ C ₆ ; ¹⁵ N]glucosamine hydrochloride)	Omicron Biochemicals	Cat# GLC-094
N-acetyl-D-[UL- ¹³ C ₆ ; ¹⁵ N]glucosamine	Omicron Biochemicals	Cat# GLC-051
d ₆ -Spermidine	MedChemExpress	Cat# HY-B1776S
[¹³ C ₄]Putrescine ([¹³ C ₄]1,4-Butanediamine)	Cambridge Isotope Laboratories	Cat# CLM-6574-0.1
2-Acetamido-2-deoxy-β-D-glucopyranosyl-L-asparagine	Biosynth	Cat# MA09584
d ₇ -ADMA	Cambridge Isotope Laboratories	Cat # DLM-7476-PK
[1,2- ¹³ C ₂]L-arginine:HCL	Cambridge Isotope Laboratories	Cat# CLM-2051-0.1
ADMA dihydrochloride	MedChem Express	Cat# HY-113216
L-lysine hydrate	Sigma Aldrich	Cat# 282677
N _ε ,N _ε ,N _ε -Trimethyllysine hydrochloride	Santa Cruz Biotechnology	Cat #sc-215597
choline bitartrate	Sigma-Aldrich	Cat# C1629

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
L-arginine	Sigma Aldrich	Cat #A5006
Envigo T.2019 Teklad Global 19% Protein Extruded Rodent Diet	Inotiv	Cat# T.2019
RPMI 1640 Culture	GIBCO	Cat# 11875
L-carnitine inner salt	Sigma	Cat# C0158
ECL	Perkin Elmer #NEL105001EA	Cat# #NEL105001EA
ECL	Cytiva	Cat# RPN2232
Critical commercial assays		
RNeasy Mini Kit	Qiagen	Cat #74104
Deposited data		
RNAseq	This paper, GEO	GEO 308565
Uncropped western blots and raw values used to generate all graphs	This Paper	Data S1 - Source Data
Code	This paper, Zenodo	Zenodo https://doi.org/10.5281/zenodo.16067534
LC-MS Files	This paper, Metabolomics Workbench	Metabolomics Workbench ⁴⁵ (PR002544).
Experimental models: Cell lines		
HeLa	ATCC	Cat# CCL-2
HEK293T	ATCC	CRL-1573
Jurkat	ATCC	TIB-152
Experimental models: Organisms/strains		
C57BL/6J mice	The Jackson Laboratory	000664
B6.Cg-Gt(ROSA)26Sor ^{tm1(CAG-EGFP)Br/y} /J	The Jackson Laboratory	032290; RRID: IMSR_JAX:032290
SLC25A45 KO mice	This paper	N/A
Oligonucleotides		
Oligonucleotides	IDT	This paper; sequences in Table S2
Recombinant DNA		
lentiCRISPR v2	Addgene	Cat# 52961; RRID: Addgene_52961
lentiCRISPR v1 (GFP)	Addgene	Cat# 49535; RRID: Addgene_49535
pLV-EF1a-IRES-Puro	Addgene	Cat# 85132; RRID: Addgene_85132
pLV-EF1a-IRES-Blast	Addgene	Cat# 85133; RRID: Addgene_85133
pMXs-3×HA-EGFP-OMP25	Addgene	Cat# 83356; RRID: Addgene_83356
pMXs-3×Myc-EGFP-OMP25	Addgene	Cat# 83355; RRID: Addgene_83355
Δ-VPR packaging plasmid	Addgene	Cat# 8454; RRID: Addgene_8454
CMV VSV-G	Addgene	Cat# 8455; RRID: Addgene_8455
pLV-EF1a-IRES-Blast 3XHA-mCherry-OMP25	This paper	N/A
pLV-EF1a-IRES-Blast 3XMyc-mCherry-OMP25	This paper	N/A
pLV-EF1a-IRES-Blast 3xFlag-SLC45A4	This paper	N/A
pLV-EF1a-IRES-Blast BBOX1-3xFLAG	This paper	N/A
pLenti v2 Puro SLC25A45	This paper	N/A
pLenti v2 Puro SLC45A4	This paper	N/A
pLenti v1 GFP SLC25A45	This paper	N/A
pLV-EF1a-IRES-Blast SLC25A45 R283A	This paper	N/A
pLV-EF1a-IRES-Blast SLC25A45 R283C	This paper	N/A
pLV-EF1a-IRES-Blast Asp121Ala,	This paper	N/A
pLV-EF1a-IRES-Blast Lys124Ala,	This paper	N/A
Software and algorithms		
Skyline Daily v22	MacCoss Lab	https://skyline.ms/project/home/software/Skyline/begin.view ; RRID: SCR_014080

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Prism	GraphPad	https://www.graphpad.com/scientific-software/prism ; RRID: SCR_002798
Code	This paper	Zenodo: https://github.com/artemkhan/SLC_ML
Code	This paper	Github https://doi.org/10.5281/zenodo.16067534

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

Slc25a45 KO mice were generated by the CRISPR/Genome Editing and the Transgenic/Reproductive Center at Rockefeller University. C57BL/6J (000664) mice were obtained from The Jackson Laboratory. All animal procedures were performed under the guidelines of the Institutional Animal Use and Care Committee (IACUC) at Rockefeller University. Prior to the experiments, the mice were subjected to two weeks of a vegetarian diet (Envigo T.2019 Teklad Global 19% Protein Extruded Rodent Diet).

METHOD DETAILS

Generation of the machine learning prediction pipeline

GWAS of plasma metabolites

We previously conducted the largest-to-date TWAS of the human metabolome.⁴ We utilized two large-scale blood metabolomic studies, leveraging GWAS summary statistics from the METSIM study (1391 metabolites in 6136 Finnish men)⁸ and the CLSA cohort⁹ (1,091 metabolite and 309 ratios, 8,299 individuals of European ancestry). Harmonization of metabolites was performed according to the study by Chen et al.⁹

Building the training dataset

We parsed the Human cyclopedic gene-metabolite associations⁴⁶ into gene-metabolite associations. For the training dataset, we removed the associations that are not coming from the free metabolites (i.e., removing the compounds involved in the reactions bound to DNA, tRNA, protein, histone modifications, etc.) given they are unlikely to have a global effect on plasma metabolite levels. Additionally, we filtered out the frequent flyer metabolites that have associations with 30 or more genes and are likely to be non-specific cofactors to the considered protein.

Building the reaction and chemical similarity map

For the reaction map, we considered all the associations including the ones that are not acting on a free metabolite as it will still be able to provide information on the mapping between the metabolites. We removed the associations from the map involving the metabolite that had over 20% of metabolites as neighbors to avoid inflating the significance. For the chemical similarity map, we relied on the name similarity of the metabolites: they should be either a substring of each other or share a common root. To build the dictionary of the roots, we identified the longest common substring between each pair of the metabolites. Then, we filtered out the entries that are substrings of more than 50 metabolites and manually curated the list.

Minimum GWAS and TWAS *p* Values

CLSA and METSIM datasets were used for the analysis as indicated in the GeneMAP work.⁴ We defined the boundaries based on ensemble 98 ± 15 Kb pairs for the gene and considered only protein coding genes. For TWAS, we utilized the minimum *p*-value across the expression model as in the GeneMAP work.⁴

Pathogenicity score

The pathogenicity score was defined based on the gene mapping from CADD scores as described in the GeneMAP work.⁴

Building the neighbors by metabolite reaction mapping

We searched for the genes in the reaction map that are associated with the considered metabolite. Afterward, we generated a list of all the metabolites that are connected to the identified genes excluding the considered metabolite. Finally, we considered all the association of the gene in question with the generated list of metabolites and computed minimum GWAS and TWAS *p*-values across these entries.

Building the neighbors by the chemical mapping

We considered all the association of the gene in question with the generated list of metabolites as outlined above. Then, we computed minimum GWAS and TWAS *p*-values across these entries excluding the considered metabolite.

Building the feature matrix

We created a matrix with 17 features that include the CADD pathogenicity score for the gene of interest in both the CLSA and METSIM datasets. The matrix also includes the minimum TWAS and GWAS *p*-values for the gene-metabolite associations, as well as for the gene's reaction and chemical network neighbors. Additionally, it lists the number of CLSA and METSIM gene-metabolite associations considered based on the reaction and chemical networks. Only pairs with all the entries present have been considered. Of note, supervised labels—experimentally validated gene-metabolite pairs—were compiled independently of all feature sets sources

(e.g., GWAS summary statistics, CADD) to limit the chances of data leakage. The summary of the features used is provided in the Table S1.

Model choice

The key python packages have been used are tensorflow⁴⁷ and sklearn.⁴⁸ The dataset was divided into training and validation sets using k-fold cross-validation ($k = 5$), ensuring that each fold provided a balanced evaluation of model performance (the labels and features are discussed above). Within each fold, missing values in the feature columns were imputed using the median value calculated from the training data.

Neural network training

To standardize the feature values, we applied Z score normalization to the training data, where each feature was centered to have a mean of zero and scaled to a unit variance. The same scaling factors (mean and standard deviation) derived from the training data were then applied to the validation data to avoid any bias that could occur if validation data influenced the scaling process. A fully connected neural network was implemented to classify the target variable. The network architecture consisted of an input layer matching the number of features in the dataset, followed by two hidden layers with ReLU activation. A grid search over hyperparameters was conducted for the number of units in each layer (32, 64; 16, 32), dropout rates (0.2–0.5), learning rates (0.001–0.01), batch size (fixed to 32), and the number of epochs (max = 50). Dropout was incorporated into the architecture to reduce overfitting by randomly setting a fraction of input units to zero at each update during training. The model was compiled using the Adam optimizer with binary cross-entropy as the loss function. Early stopping was implemented to prevent overfitting, with training halted if the validation loss did not improve after five consecutive epochs. The neural network was trained on the scaled training data and validated using the scaled validation data for each fold.

Random forest model

A Random Forest classifier was trained using default hyperparameters, with 200 estimators, a maximum tree depth of 10, and a minimum of 5 samples required to split an internal node. The model was trained on the original, unscaled features, and predictions were made on the validation set. To further evaluate model performance, we computed the AUC. The receiver operating characteristic (ROC) curve was also generated for the Random Forest classifier, and a specific threshold corresponding to a desired false positive rate (FPR = 0.05) was identified. This threshold was used to adjust the classification threshold to control for the false positive rate.

Gradient boosting model

We also implemented a Gradient Boosting Classifier using the default hyperparameters to assess its performance on the dataset. This model iteratively trains weak learners (decision trees) and improves upon the performance of previously trained models by minimizing the loss function.

Support vector machine (SVM) model

Finally, a Support Vector Machine (SVM) model with a radial basis function (RBF) kernel was trained on the scaled training data.

Model evaluation

For each model and fold, performance was recorded using AUC. These results were computed for each fold to provide a robust comparison of model performance.

Predictions

We trained the Random Forest model on the whole training datasets (the labels and features are as indicated above) and made predictions on the unseen dataset. The classifier is a model that outputs a single probability score ranging from 0 to 1 for each gene-metabolite pair, with a higher score (closer to 1) indicating a greater likelihood of a true gene-metabolite association. Gene-level outputs were derived, as described above, from GWAS and TWAS gene-metabolite associations.

Prioritization criteria

First, we identified associations with the probability score >0.697775 , which corresponds approximately to FDR of 5% as estimated from the Random Forest probability score threshold in the training dataset. We ranked the obtained associations gene-metabolite association by the gene or by metabolite. To minimize the effect of the frequent flyer metabolites and genes, we applied the following criteria: (a) the metabolite should have a rank of <10 for the considered gene; (b) the gene must be either in top 50% percent for the given metabolite (if less than 50 genes are associated with metabolite) or among top 500 genes. After we applied the following filters, we focused on solute carriers given our interest in this class of transporters. The list has been defined according as indicated in Khan et al.¹ The entries from Figure S2A are provided in Table S2.

SLC-metabolite network analysis

We utilized the prioritized SLC-metabolite associations to construct edges; SLCs and metabolites were used as nodes. To identify biologically interpretable clusters, we performed Louvain community detection.

Cell lines and reagents

Human cell line HEK293T, Jurkat, HeLa were purchased from the ATCC. The cell lines were confirmed to be free of mycoplasma contamination. The identities were verified by STR profiling. The cells were cultured in RPMI 1640 medium (Gibco) containing 2mM glutamine, 10% FBS, 1X penicillin and streptomycin (Gibco, #15140-122). Antibodies against HA-Tag (3724S, 1:200 for immunofluorescence), citrate synthase (14309S, 1:1,000 for Western blot) were from Cell Signaling Technology; SLC25A12 (ab200201, 1:1,000 for Western blot) was from Abcam; GAPDH (GTX627408 1:1,000 for Western blot) was from GeneTex, FLAG Tag (F1804, 1:2,000 for Western blot, 1:200 for immunofluorescence) was from Sigma-Aldrich, SLC25A45 (PA5-62251, 1:1,000 for Western blot, 1:50 for immunofluorescence) was from Invitrogen, ATP5A1 (439800, 1:200 for immunofluorescence) was from Thermo Fisher

Scientific, LAMP-1 (1D4B, 1:1000 for Western blot) was from Developmental Studies Hybridoma Bank, Calreticulin (GTX83089, 1:1,000 for Western blot) was from GeneTex. Anti-mouse IgG–HRP (7076S, 1:3,000 for Western blot), anti-rabbit IgG–HRP (7074S, 1:3,000 for Western blot), anti-rat IgG–HRP (7077S, 1:3,000 for Western blot) were obtained from Cell Signaling Technology. Antibodies for immunofluorescence staining were goat anti-mouse Alexa Fluor 488 (A11029, 1:300), goat anti-rabbit Alexa Fluor 555 (A21428, 1:300), donkey anti-rabbit Alexa Fluor 488 (A21206, 1:300), or donkey anti-mouse Alexa Fluor 555 (A10037, 1:300) from Thermo Fischer Scientific. Other reagents: anti-HA magnetic beads (88837, Thermo Scientific Pierce); DAPI (D1306, ThermoFisher Scientific); polybrene (H9268), puromycin (P8833) (Sigma); blasticidin (ant-bl-1, Invivogen); Phalloidin-iFluor 647 Conjugate (Cat No. 20555, Cayman Chemical Company Item).

Generation of knockout and cDNA-overexpression cell lines

Oligos for cloning *SLC25A45* and *SLC45A4* sgRNA were synthesized by IDT and cloned into BsmBI linearized lentiCRISPR-v2 plasmid (Addgene, #75159) with T4 ligase (NEB #M0202). cDNA for *SLC25A45* as well as *SLC25A45* mutants, *UNC93A-3xFlag*, or *3xFlag-SLC45A4* were cloned into BamHI–HF (NEB #R3136L) and EcoRI–HF (NEB #3101L) linearized pLV–EF1a–IRES–Puro (Addgene, #85132) plasmid by Gibson assembly. Mito-tag plasmids were based on pMXs–3XHA–EGFP–OMP25 (Addgene, #83356) and pMXs–3XMyC–EGFP–OMP25 (Addgene, #83355) with mCherry being replaced for the selection. The cDNAs for *3XHA-mCherry-OMP25* and *3XMyC-mCherry-OMP25* were cloned into pLV–EF1a–IRES–Blast backbone (Addgene, #85133) by using Gibson assembly. sgRNA- or cDNA-expressing vectors and lentiviral packaging vectors Delta–VPR and CMV VSV–G were transfected into HEK293T cells using XTremeGene 9 transfection reagent (Roche #636478700). The virus-containing supernatant was collected 48 h after transfection and passed through a 0.45- μ m filter. The transduction of target cells was performed in 6-well tissue culture plates by addition to the medium collected virus and 4 μ g mL^{−1} of polybrene, followed by centrifugation at 1,126g for 80 min. The next day, the media was changed to remove the virus. *SLC25A45* KO cells were selected by fluorescence associated cell sorting (FACS) of the GFP+ cells (lentiCRISPR–v1(GFP) and consequent single-cell cloning (when appropriate) or by puromycin selection and consequent FACS-based single-cell cloning (when appropriate). HEK293T *SLC25A45* and *SLC45A4* KO cells single clones have been generated by the transient transfection, followed by puromycin selection and FACS-based single cell cloning. For Mito-tag cells, mCherry+ cells were selected by FACS (pLV–3XHA–mCherry–OMP25, pLV–3XMYC–mCherry–OMP25). As no validated antibody is available for a reliable endogenous detection, sg*SLC25A45* and sg*SLC45A4* transduced/transfected cells were tested by ICE analysis (Synthego) of genomic DNA Sanger sequencing. Cells expressing *SLC25A45* and *UNC93A-3xFLAG* were selected with puromycin. Cells expressing *BBOX1-3xFLAG*, *3xFLAG-SLC45A4* were selected with blasticidin. For all the mentioned constructs the matching empty vectors (without insert) were used as a control. We validated all constructs by Sanger sequencing. The nucleotide sequences are provided in Table S2.

Immunofluorescence staining and confocal microscopy for co-localization

Sterile coverslips were coated with 50 μ g/mL Poly–D–Lysine (ChemCruz #sc–136156) for 30 min at 37°C. Subsequently, 500,000 cells were seeded per well in a 6-well plate and cultured overnight. The following day, cells were fixed with 4% PFA for 15 min, permeabilized using 1% Triton X–100 in PBS, blocked with 1% bovine serum albumin (Sigma) for 30 min, and then incubated with the primary antibody overnight at 4°C. The next day, cells were washed twice with PBS and incubated with secondary antibodies for 30 min. After two additional PBS washes, the coverslips were incubated with DAPI for 10 min, washed twice in PBS, and mounted using ProLong Gold antifade mountant (Molecular Probes). Imaging was performed using a Nikon A1R MP multiphoton microscope in confocal mode with a Nikon Plan Apo γ 60X/1.40 oil immersion objective.

Immunoblotting

Transmembrane buffer (10mM Tris–HCl pH 7.4, 150mM NaCl, 1mM EDTA, 1% Triton X–100, 2% SDS, 0.1% CHAPS) supplemented with protease inhibitors (EMD Millipore #535140 or Sigma–Aldrich #11836170001) was used to lyse the cells, followed by sonication. For mitochondrial immunopurification experiments, 1% Triton X–100 Buffer (50mM Tris–HCl pH 7.4, 150mM NaCl, 1mM EDTA, 1% Triton X–100) supplemented with protease inhibitors (Sigma–Aldrich #11836170001) was used for lysis. Lysates were centrifuged at maximum speed (>14,000g) at 4°C, and supernatant was collected. Protein quantification was performed by using Pierce BCA Protein Assay Kit (Thermo Fisher #23227) with bovine serum albumin as a standard. 10–20% Tris–Glycine gels (Invitrogen #XP10205BOX) were used to resolve the samples. Transfer to PVDF membrane (EMD Millipore #1PVH00010) was performed with CAPS buffer (10mM CAPS, 10% ethanol). Membrane was blocked in skim milk (5% w/v) or bovine serum albumin (5% w/v) and incubated in primary antibodies at 4°C overnight. Secondary antibody incubation was in anti-mouse IgG–HRP (CST #7076), anti-rabbit IgG–HRP (CST #7074), or anti-rat IgG–HRP (CST #7077) diluted at 1:3000 ratio in skim milk or bovine serum albumin. Washes prior to secondary antibody incubation and blot development were conducted with 0.1% Tween 20 Tris–buffered saline. ECL chemiluminescence was used (PerkinElmer #NEL105001EA or Cytiva #RPN2232) with autoradiography films (Thomas Scientific #1141J52) and SRX–101A Film Processor (Konica Minolta).

Whole cell metabolomics experiments

500,000 HEK293T cells were seeded per well in 6-well plates in triplicate. The following day, media was changed to the RPMI containing an indicated dose of the tracer (*d*₉–TML (Toronto Chemical Cat #T796447), 2–[¹⁵N]amino–2–deoxy–D–[UL–¹³C₆]glucose hydrochloride (D–[UL–¹³C₆; ¹⁵N]glucosamine hydrochloride) (Omicron Biochemicals, GLC–094), N–acetyl–D–[UL–¹³C₆; ¹⁵N]glucosamine

(Omicron Biochemicals, GLC-051), d_6 -spermidine (MedChem Express, HYB1776S), [$^{13}\text{C}_4$]putrescine ([$^{13}\text{C}_4$]1,4-Butanediamine) (Cambridge Isotope Laboratories, CLM-6574-0.1)) or compound of interest (2-Acetamido-2-deoxy-b-D-glucopyranosyl L-asparagine, Biosynth MA09584). For the spermidine and putrescine, the tracing has been performed in HBSS for the cell profiling and serum free RPMI for the media profiling. At indicated timepoints, cells were washed twice with cold 0.9% NaCl (w/v), followed by metabolite extraction in ice-cold 80:20 LC/MS grade methanol: water buffer containing ^{15}N and ^{13}C fully-labeled amino acid standards (Cambridge Isotope Laboratories #MSK-A2-1.2). Samples were vortexed for 10 min at 4°C and then centrifuged at maximum speed ($>14,000g$) at 4°C for 10 min. Supernatants were dried under nitrogen, stored at -80°C , and resuspended in 50% Acetonitrile before injection onto LC-MS. For the media profiling, 10 μL of the media supernatant was added to 90 μL of the cold 75% acetonitrile:25% methanol buffer containing heavy amino acid standards, mixed, centrifuged for 10 min at maximum speed ($>14,000g$) at 4°C, and stored at -80°C .

Polar metabolite LC-MS profiling

Metabolite analysis was conducted using an Orbitrap IQ-X Tribrid mass spectrometer or a Q Exactive Plus Orbitrap mass spectrometer. Both MS were coupled to a Vanquish Horizon UHPLC. For the general profiling samples run on Q Exactive Plus, for separation, polar metabolites were separated by HILIC chromatography (ZIC-pHILIC $150 \times 2.1\text{ mm}$ column, EMD Millipore) at 150 $\mu\text{L}/\text{min}$ with a gradient from 90% B to 40% B over 30 min (A: 20 mM ammonium carbonate in water, pH 9.3; B: 100% acetonitrile). Solvents were LC/MS grade (Optima, ThermoFisher). Samples were analyzed in both positive and negative mode using polarity switching. The HESI source was operated with a spray voltage of 3000V in both positive and negative modes. The capillary temperature was set to 275°C , with sheath gas at 40 and aux gas at 15. The scan parameters were MS1 resolution of 140,000, AGC target of $1\text{e}6$, and maximum IT of 80 ms. The scan range used was 55–825 m/z . For the tracing samples run on Q Exactive Plus, Polar metabolites were separated by HILIC chromatography (ZIC-pHILIC $150 \times 2.1\text{ mm}$ column, EMD Millipore) at 150 $\mu\text{L}/\text{min}$ with a gradient from 90% B to 40% B over 30 min (A: 20 mM ammonium carbonate in water, pH 9.3; B: 100% acetonitrile). Solvents were LC/MS grade (Optima, ThermoFisher). Samples were analyzed in both positive and negative mode using polarity switching. The HESI source was operated with a spray voltage of 3000V in both positive and negative modes. The capillary temperature was set to 275°C , with sheath gas at 40 and aux gas at 15. The scan parameters were MS1 resolution of 140,000, AGC target of $1\text{e}6$, and maximum IT of 80 ms. The scan range used was 55–825 m/z . For the analysis run on IQ-X, samples were loaded on a SeQuant ZIC-pHILIC column ($150 \times 2.1\text{ mm}$, 5- μm polymeric, EMD Millipore) and a SeQuant ZIC-pHILIC Guard Kit ($20 \times 2.1\text{ mm}$) at 40°C and eluted with a solvent system composed of mobile phase A (20 mM ammonium carbonate and 0.1% ammonium hydroxide in water at pH 9.3) and mobile phase B (100% acetonitrile). The injection volume was set to 5 μL , and samples were maintained at 4°C . The gradient (vol/vol) used was as follows: 0–22 min, linear gradient from 90% to 40% B; 22–24 min, held at 40% B; 24–24.1 min, returned to 90% B; 24.1–30 min, equilibrated at 90% B at a flow rate of 150 $\mu\text{L}/\text{min}$. The IQ-X Tribrid was operated in full-scan polarity-switching mode with a spray voltage of +3 kV and -2.5 kV , sheath gas at 40, auxiliary gas at 15, sweep gas at 1, an ion-transfer tube temperature of 275°C , and a probe heater temperature of 350°C . Samples were analyzed with a resolution of 120,000, an AGC target of 1.2×10^5 , a maximum injection time of 246 ms, and two mass ranges of 55–400 Th and 375–875 Th. Both mass spectrometers were externally calibrated every three days or weekly using Pierce Calibration Solution (Thermo Fisher Scientific). Sample injections were randomized.

LC/MS analysis for lipidomics

Ultra-high performance liquid chromatography coupled with mass spectrometry (UHPLC/MS) analyses were conducted using a Thermo Scientific Vanquish Flex UHPLC system, interfaced with a Thermo Scientific Orbitrap ID-X Mass Spectrometer. For the separation of lipids, an Acquity Premier HSS T3 column ($2.1 \times 100\text{ mm}$, $1.8\mu\text{m}$) was utilized. The mobile-phase solvents consisted of solvent A = 10mM ammonium formate, 5 μM medronic acid in 5:3:1 water:acetonitrile:2-propanol and solvent B = 10 mM ammonium formate in 1:9:90 water:acetonitrile:2-propanol. The column compartment temperature was maintained at 55°C and lipids were eluted using a linear gradient at a flow rate of 400 $\mu\text{L}/\text{min}$ as follows: 0 min, 15% B; 2.5 min, 50% B; 2.6 min, 57% B; 9 min, 70% B; 9.1 min, 93% B; 11 min, 96% B; 11.1, 100% B; 11.1–12min, 100% B; 12.2 min, 15% B; 12.2–16 min, 15% B. Data was acquired in positive and negative ion modes. The LC/MS data were then processed and analyzed using XCMS,⁴⁹ CompoundDiscoverer, and Skyline.⁵⁰

Immunopurified mitochondria metabolic profiling from the cells and mouse tissues

Mitochondrial immunopurification from HEK293T, Jurkat or HeLa cells expressing either 3 $\times$ HA-OMP25-mCherry (immunopurified mitochondria) or 3 $\times$ Myc-OMP25-mCherry (background control) and from liver and kidney of the MitoTag 7-week old *Slc25a45*-knockout and wild type female mice as well as control (no MITO Tag) wild type mouse was conducted according to the protocol by Chen et al.³² and Bayraktar et al.³¹ with slight modifications. Prior to the collection of serum and harvesting the organs, mice were fasted for 4 h. Cells were washed in a confluent 15 cm dish twice with cold saline (0.9% NaCl), collected in cold KPBS, and centrifuged at 1,000g for 1.5 minutes at 4°C . The pellet was then resuspended in 1 mL of KPBS, homogenized with 30 strokes in a 2-mL homogenizer and centrifuged at 1,000g for 1.5 minutes at 4°C . For the mouse tissues, following the collection liver and kidney, we homogenized them in 1 mL of KPBS and centrifuged at 1,000g for 1.5 minutes at 4°C . Part of the supernatant was taken for the whole cell/tissue protein lysis and metabolic extraction. The remaining supernatant was incubated with 200 μL of anti-HA magnetic beads on a rotator shaker for 5 minutes at 4°C . Following the incubation, the beads were washed three times with ice-cold KPBS.

Consequently, 10% of the bead volume was used for lysis with 1% Triton buffer (mitochondrial protein sample). The remaining volume was extracted in 80% methanol containing heavy isotope labeled amino acid standards. The protein and metabolite samples were placed on a rotor for 10 min at 4°C, followed by 10 min centrifugation at maximum speed (>14,000g) at 4°C. Samples for the following metabolomic analysis were stored at –80°C.

Isotope labeled uptake in immunopurified mitochondria

Mitochondrial Immunopurification was performed according to the protocol outlined above. Immunopurified mitochondria were incubated in the mitochondrial uptake buffer for indicated timepoints at room temperature (KPBS, 10mM HEPES, 0.5mM EGTA, 25 μ M d₉-TML (Toronto Chemical Cat #T796447), 25 μ M d₇-ADMA (Cambridge Isotope Laboratories Cat # DLM-7476-PK), pH~7.35), indicated dose of [1,2-¹³C₂]L-arginine:HCL (Cambridge Isotope Laboratories, CLM-2051-0.1). For the competitive uptake with unlabeled substrate, mitochondria were incubated in the following mitochondrial uptake buffer: KPBS, 10mM HEPES, 0.5mM EGTA, 10 μ M d₉-TML (Toronto Chemical, Cat #T796447), 500 μ M unlabeled ADMA dihydrochloride (MedChem Express, HY-113216), L-lysine hydrate (Sigma Aldrich, 282677), N_ε,N_ε,N_ε-Trimethyllysine hydrochloride (Santa Cruz Biotechnology, sc-215597), choline bitartrate (Sigma-Aldrich, C1629) and L-arginine (Sigma Aldrich, A5006). The pH of the buffer was adjusted to ~7.35. Uptake was quenched by ice-cold KPBS, followed by three washes in cold KPBS, and extracted in 80% methanol. The sample was then placed on a rotor for 10 min at 4°C, centrifuged for 10 min centrifugation at maximum speed (>14,000g) at 4°C, and stored at –80°C.

Mouse work

Details

Slc25a45 KO mice were generated by injecting the Cas9/gRNA RNP complex into C57Black/6J mouse zygotes to create indels in exon 4. A gRNA binding to genomic target (5'- GCTGTACACTCCGAACAGGACGG) on exon 4 of Slc25a45 was selected based on the evaluation of off-target potential using the web-based tool CRISPOR (<http://crispor.tefor.net>) and of on-target efficiency in mouse embryos. We used the synthetic crRNA and tracrRNA from IDT (Integrated DNA Technology) to assemble the gRNA. The preparation of a microinjection cocktail with gRNA and Cas9 protein was described previously (see reference below). Briefly, crRNA and tracrRNA were annealed by heating to 98°C for 2 min and cooling at room temperature for 5 min. Cas9 protein (0.2 μ M, Cat#1081061, IDT) was added to the solution containing the assembled gRNA (0.1 μ M), incubated at 37°C for 15 min to form the ribonucleoprotein complex, purified by Millipore Ultrafree column (Cat# UFC30VV25, Millipore), and placed on ice until microinjection. After microinjection, embryos were transferred to the oviduct of surrogate mice on the same day and allowed to develop to term.⁵¹ Slc25a45 KO mice were then bred to MITO-TAG mice B6.Cg-Gt(ROSA)26Sortm1(CAG-EGFP)BrSy/J.³¹

Mouse serum for metabolomic analysis

Mouse blood was collected in EDTA-containing tube and centrifuged at 500g for 5 min at 4°C. 5 μ L of the supernatant was added to 45 μ L of the cold 75% acetonitrile:25% methanol buffer containing heavy amino acid standards, mixed, centrifuged for 10 min at maximum speed (>14,000g) at 4°C, and stored at –80°C.

Tissues for polar metabolomic and lipidomic analysis

Tissue samples stored at –80°C were placed onto dry ice alongside pre-chilled 2 mL reinforced microtubes containing 2.8 mm ceramic beads (Omni Cat#.19–628). Samples were transferred into the microtubes, and 2:2:1 methanol:acetonitrile:water (MAH; LC-MS Grade) at 40 μ L per mg of tissue (minus 100 μ L) was added to the tube. Samples were immediately homogenized using the Omni Bead Ruptor Elite Homogenizer (speed: 6 m/s; run time: 20 s) equipped with a cooling hood attachment and the Omni BR-Cryo Cooling Unit at medium flow. Three cycles of homogenization were performed, with tubes cooled between cycles. Post-homogenization, samples were transferred into labeled 1.7 mL Eppendorf tubes, followed by centrifugation at maximum speed (>12,800 rcf) for 10 min at 4°C. The supernatant was transferred carefully to new Eppendorf tubes without disturbing the pellet. Pellets were resuspended in 40 μ L isopropanol (IPA; LC-MS Grade) per mg tissue (minus 100 μ L), mixed thoroughly, and homogenized as previously described. Following identical incubation and centrifugation steps, the supernatant was transferred into separate Eppendorf tubes. Extracts for lipidomics were prepared by mixing MAH and IPA in a 1:1 ratio. The samples were stored at –80°C.

In vivo d₉-TML tracing

The experiment was performed in *Slc25a45*^{–/–} vs. *Slc25a45*^{+/-}, *Slc25a45*^{+/+} 6-7-week-old female mice. On the day of the experiment, mice were administered 100 μ L of 100 μ M d₉-TML solution in PBS (Toronto Chemical, Cat #T796447) via tail vein injection. Serum, liver, and kidney were harvested an hour post-injection and processed for the LC-MS analysis as indicated above. For the experiment on wild type mice (Jackson laboratory), mice were administered 100 μ L of 200 μ M d₉-TML solution in PBS (Toronto Chemical, Cat #T796447) via tail vein injection. Serum was harvested an hour post-injection and processed for the LC-MS analysis as indicated above.

Fasting experiments

7-week-old male *Slc25a45*^{–/–} vs. *Slc25a45*^{+/-}, *Slc25a45*^{+/+} mice were transferred into a clean cage with or without the food access for 24 h. At the beginning of experiment as well as 24 h after, the blood was collected for serum analysis. Subsequently, the liver

tissues were harvested, immediately flash frozen and stored at -80°C for subsequent metabolite profiling by liquid chromatography-mass spectrometry (LC-MS). For the fasting experiment with carnitine supplementation, 6- to 8-week-old female *Slc25a45*^{-/-} vs. *Slc25a45*^{+/-}, *Slc25a45*^{+/+} mice were pre-treated for 72 h with either regular drinking water or water supplemented with 10 g/L L-carnitine inner salt (Sigma-Aldrich, C0158). At the start of the fasting experiment, mice on regular water were transferred to clean cages and either (a) maintained with *ad libitum* food access or (b) subjected to a 24-h fast with continued access to regular drinking water. Mice receiving carnitine supplementation were transferred to clean cages and fasted for 24 h with continued access to drinking water containing 10 g/L L-carnitine inner salt (Sigma-Aldrich, C0158).

RNA-sequencing

Total RNA was extracted from the livers harvested as described above from *Slc25a45*^{-/-} vs. *Slc25a45*^{+/-}, *Slc25a45*^{+/+} mice using the RNeasy Mini Kit (Qiagen #74104) in accordance with the manufacturer's protocol. Library preparation and RNA-sequencing was performed by Genewiz. The analysis of the sequencing data was performed as in Timson et al. with some modifications.⁵² Quality control of the sequencing data was conducted with Rfastp.⁵³ The reads alignment was performed according to the reference genome (Mus_musculus.GRCm39.cdna.all.fa.gz), and gene models were based on Mus_musculus.GRCm39.107.gtf.gz. Transcript expression was computed using Salmon,⁵⁴ followed by processing with DESeq2,⁵⁵ with the interaction model design = ~ genotype + fasting + genotype:fasting. The interaction term (genotype:fasting) specifically tests whether the effect of fasting on gene expression is significantly different in the knockout mice compared to wild-type. Gene Set Enrichment Analysis (GSEA) was applied to genes ranked by the interaction statistic to assess pathway-level differential responses to fasting between genotypes. The terms for Gene Set Enrichment (GSEA) and Gene Ontology (GO) analysis were retrieved from Mouse MSigDB Collections Gene Ontology Analysis.^{55–57} GSEA was performed using the fgsea() package.⁴⁵

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis and data visualization

Statistical analysis and data visualization were performed in Prism9 (GraphPad Software), R (4.3.1), or ChimeraX-1.7.⁵⁸ The details of statistical analysis for each experiment are described in the figure legends.